Modular Forms

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1 The Upper-half Plane and Congruence Subgroups

If a is an integer modulo m with $(a,m)=1$ we write a^{-1} for the inverse of a modulo m. For any complex z let

$$e(z)=e^{2\pi iz}.$$

We will distinguish the primitive third root of unity $\omega=e\left(\frac{1}{3}\right)$. Also, all sums and products over a quotient group will be consider to be over a complete set of representatives.

1.1 The Upper-half Plane

The group $\mathrm{GL}_2^+(\mathbb{R})$ acts on the Riemann sphere $\widehat{\mathbb{C}}$ by Möbius transformations. Explicitly, any $\alpha=\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\in\mathrm{GL}_2^+(\mathbb{R})$ acts on $z\in\widehat{\mathbb{C}}$ by

$$\alpha z=\frac{az+b}{cz+d}.$$

Moreover, this group action is by automorphisms of the Riemann sphere and is invariant under scalar multiplication as well as negation. A short computation shows

$$\operatorname{Im}(\alpha z) = \det(\alpha)\frac{\operatorname{Im}(z)}{|cz+d|^2}, \tag{1}$$

Therefore α preserves the sign of the imaginary part of z. This means $\mathrm{GL}_2^+(\mathbb{R})$ preserves the upper half-plane $\mathbb{H}$, the lower half-plane $\overline{\mathbb{H}}$, and the extended real line $\widehat{\mathbb{R}}$. Of course, the subgroup $\mathrm{GL}_2^+(\mathbb{Q})$ also preserves $\widehat{\mathbb{Q}}$.

We will be primarily interested in this action on the upper-half plane $\mathbb{H}$. We equip $\mathbb{H}$ with the **hyperbolic measure** $d\mu$ given by

$$d\mu = \frac{dx\,dy}{y^2}.$$

This is simply the Lebesgue measure on $\mathbb{C}$ restricted to $\mathbb{H}$ and scaled by a factor of $\frac{1}{y^2}$. The most important property about the hyperbolic measure is that it is $\mathrm{GL}_2^+(\mathbb{R})$-invariant.

Proposition 1.1. *The hyperbolic measure is* $\mathrm{GL}_2^+(\mathbb{R})$*-invariant.*

Proof. Let $\alpha = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{R})$. Since αz is a Möbius transformation it satisfies the Cauchy-Riemann equations. So for the change of variables $z \mapsto \alpha z$, the absolute value of the Jacobian determinant is the absolute value of the square of the derivative of αz. A short computation shows that this is $\frac{\det(\alpha)^2}{|cz+d|^4}$. From this fact and Equation (1), a short computation shows

$$\mu(\alpha z) = \mu(z).$$

This means $d\mu$ is $\mathrm{GL}_2^+(\mathbb{R})$-invariant. □

1.2 Congruence Subgroups

To define congruence subgroups, we first need to introduce an associated group. We call $\mathrm{SL}_2(\mathbb{Z})$ the **modular group** which is the group of matrices with integral entries and of determinant 1. We will distinguish two matrices

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The first matrix is of order 4 while the second has infinite order. They act on $z \in \mathbb{H}$ by

$$Sz = -\frac{1}{z} \quad \text{and} \quad Tz = z+1,$$

respectively. Therefore S acts by a pseudo-inversion while T acts by unit translation. The first result usually proved about the modular group is that it is generated by S and T.

Proposition 1.2.

$$\mathrm{SL}_2(\mathbb{Z}) = \langle S, T \rangle.$$

Proof. Let n be an integer and $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. We will prove $\gamma \in \langle S, T\rangle$ by demonstrating that it can be reduced to the identity matrix. If $c = 0$ then γ is the identity matrix since $\det(\gamma) = 1$ and we are done. So suppose $c \neq 0$ and write $a = qc + r$ for some integers q and r with $|r| < |c|$. A short computation shows

$$ST^{-q}\gamma = \begin{pmatrix} -c & -d \\ r & b - qd \end{pmatrix}.$$

The upper-left entry of this matrix strictly larger than the lower-left entry in absolute value. Therefore if we repeatedly apply this argument, it must terminate with the lower-left entry vanishing. But then we have reached the identity matrix. □

We can now introduce congruence subgroups which are special subgroups of the modular group defined by congruence conditions on their entries. We begin with principal congruence subgroups. For a positive integer N, the **principal congruence subgroup** $\Gamma(N)$ of level N is defined by

$$\Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}.$$

This congruence subgroup consists of the matrices which reduce to the identity modulo N. Whence $\Gamma(N)$ is the kernel of the homomorphism

$$\pi : \mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}) \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix} \pmod{N},$$

and is therefore a normal subgroup. A subgroup Γ of the modular group is said to be a **congruence subgroup** if it contains a principal congruence subgroup. The minimal such N for which $\Gamma(N) \subseteq \Gamma$ is called the **level** of Γ. Note that if $M \mid N$ then $\Gamma(N) \subseteq \Gamma(M)$. Thus if Γ is a congruence subgroup of level N then it contains the principal congruence subgroups $\Gamma(kN)$ for all positive integers k. This implies that congruence subgroups are preserved under intersection. In fact, if Γ_1 and Γ_2 are congruence subgroups of levels N_1 and N_2 then $\Gamma_1 \cap \Gamma_2$ is guaranteed to be a congruence subgroup of level at most $[N_1, N_2]$. Also, it turns out that the aforementioned homomorphism is surjective. This is known as **strong approximation** for the modular group.

Proposition 1.3 (Strong Approximation for the Modular Group). *Let N be a positive integer. Then the homomorphism*

$$\pi : \mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}) \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix} \pmod{N},$$

given by natural projection, is surjective.

Proof. Let U and L be the matrices in $\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$ given by

$$U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad L = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

A short computation shows that these are the images of T and TST under π respectively. So Proposition 1.2 will imply that π is surjective provided U and L generate $\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$. So let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$. We will prove $\gamma \in \langle U, L\rangle$ by demonstrating that it can be reduced to the identity matrix. To show this, observe that $U^n = U(n)$ and $L^n = L(n)$ where $U(n)$ and $L(n)$ are the upper and lower unipotent matrices

$$U(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad L(n) = \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix},$$

for n modulo N. So it suffices to show that this collection of unipotent matrices generates $\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$. If $c = 0$ then $\det(\gamma) = 1$ implies d is the inverse of a modulo N so that $\gamma = \left(\begin{smallmatrix} a & b \\ 0 & a^{-1} \end{smallmatrix}\right)$. Short computations show

$$U(-ab)\gamma = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix},$$

and

$$L(a^{-1}-1)U(1)L(a-1)U(-a^{-1}) = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}.$$

Whence γ can be reduced to the identity matrix. If $c \neq 0$ then $\det(\gamma) = 1$ and therefore c is invertible. A short computation shows

$$L(-c)U((1-a)c^{-1})\gamma = \begin{pmatrix} 1 & (d-1)c^{-1} \\ 0 & 1 \end{pmatrix},$$

and we recognize this latter matrix is as $U((d-1)c^{-1})$. Whence γ can again be reduced to the identity matrix. $\square$

As $\Gamma(N)$ is the kernel of this homomorphism, the result implies $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)] = |\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})|$. Since any congruence subgroup Γ contains a principal congruence subgroup of some level, it follows that Γ has finite index in $\mathrm{SL}_2(\mathbb{Z})$. Accordingly, we set

$$d_\Gamma = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma].$$

We now distinguish two important classes of congruence subgroups other than the principal ones. These are the congruence subgroups

$$\Gamma_1(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \pmod N \right\},$$

and

$$\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod N \right\},$$

both of which are of level N. These congruence subgroups consist of the matrices which reduce to unipotent upper triangular and upper triangular modulo N respectively. The latter subgroup $\Gamma_0(N)$ is called the **Hecke congruence subgroup** of level N. Note that $\Gamma(N) \subseteq \Gamma_1(N) \subseteq \Gamma_0(N)$.

Since the principal congruence subgroups are normal in $\mathrm{SL}_2(\mathbb{Z})$, if Γ is a congruence subgroup of level N then so is $\alpha^{-1}\Gamma\alpha$ for any $\alpha \in \mathrm{SL}_2(\mathbb{Z})$. In fact, something more general holds. Congruence subgroups are preserved under conjugation by elements of $\mathrm{GL}_2^+(\mathbb{Q})$ provided we restrict to those elements in $\mathrm{SL}_2(\mathbb{Z})$.

Lemma 1.4. *Let Γ be a congruence subgroup and $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Then $\alpha^{-1}\Gamma\alpha \cap \mathrm{SL}_2(\mathbb{Z})$ is a congruence subgroup. In particular, if $\alpha^{-1}\Gamma\alpha \subseteq \mathrm{SL}_2(\mathbb{Z})$ then $\alpha^{-1}\Gamma\alpha$ is a congruence subgroup.*

Proof. Let the level of Γ be N and choose a positive integer r such that $r\alpha, r\alpha^{-1} \in \mathrm{GL}_2^+(\mathbb{Q})$ have integral entries. Let $M = r^2N$ and notice that any $\gamma \in \Gamma(M)$ is of the form

$$\gamma = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + M \begin{pmatrix} k_1 & k_2 \\ k_3 & k_4 \end{pmatrix},$$

for integers $k_1, \dots, k_4$. Therefore $\Gamma(M) \subseteq I + M\mathrm{Mat}_2(\mathbb{Z})$ and so

$$\alpha\Gamma(M)\alpha^{-1} \subseteq I + N\mathrm{Mat}_2(\mathbb{Z}).$$

As $\Gamma(N) = (I + N\mathrm{Mat}_2(\mathbb{Z})) \cap \mathrm{SL}_2(\mathbb{Z})$, it follows that $\alpha\Gamma(M)\alpha^{-1} \subseteq \Gamma(N)$. Whence $\Gamma(M) \subseteq \alpha^{-1}\Gamma\alpha$ because Γ is of level N. Intersecting with $\mathrm{SL}_2(\mathbb{Z})$ proves the first statement. The second statement is immediate. $\square$

Note that in the case $\alpha^{-1}\Gamma\alpha \subseteq \mathrm{SL}_2(\mathbb{Z})$, normality of principal congruence subgroups force Γ and $\alpha^{-1}\Gamma\alpha$ to be of the same level. In particular, this will always happen whenever we conjugate by elements of the modular group.

1.3 Fuchsian Groups and Modular Curves

We first restrict our group action on the upper-half plane to $\mathrm{SL}_2(\mathbb{R})$. Note that for any $\alpha = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{R})$, Equation (1) simplifies to

$$\mathrm{Im}(\alpha z) = \frac{\mathrm{Im}(z)}{|cz+d|^2}. \tag{2}$$

So $\mathrm{SL}_2(\mathbb{R})$ preserves the upper half-plane $\mathbb{H}$, the lower half-plane $\overline{\mathbb{H}}$, and the extended real line $\widehat{\mathbb{R}}$. Of course, the subgroup $\mathrm{SL}_2(\mathbb{Q})$ also preserves $\widehat{\mathbb{Q}}$.

A subgroup Δ of $\mathrm{SL}_2(\mathbb{R})$ is said to be **Fuchsian** if it is discrete and acts properly discontinuously on $\mathbb{H}$. The modular group is the prototypical example of a Fuchsian group.

Proposition 1.5. *The modular group is Fuchsian.*

Proof. $\mathrm{SL}_2(\mathbb{Z})$ is discrete in $\mathrm{SL}_2(\mathbb{R})$ because $\mathbb{Z}$ is discrete in $\mathbb{R}$. So it remains to show that $\mathrm{SL}_2(\mathbb{Z})$ acts properly discontinuously on $\mathbb{H}$. To prove this, it suffices to show that for any compact subset $K \subset \mathbb{H}$ there are finitely many $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ satisfying $\gamma K \cap K \neq 0$. Set $M = \min_{z \in K} \mathrm{Im}(z)$. If $\gamma K \cap K \neq \varnothing$ then there exists a $z \in K$ such that $\mathrm{Im}(\gamma z) \geq M$. This inequality is equivalent to

$$(cx+d)^2 + (cy)^2 \leq \frac{y}{M},$$

by Equation (2), which is only satisfied for finitely many pairs of relatively prime integers (c, d) since $y > 0$. Each pair determines infinitely many matrices $\gamma_k = \left(\begin{smallmatrix} a+kc & b+kd \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ possibly satisfying $\gamma_k K \cap K \neq \varnothing$. As K is compact it has bounded real part and so only finitely many such k satisfy this condition. Hence there are finitely many possible matrices γ. $\square$

As an immediate consequence of this result, any subgroup or conjugate of the modular group is also Fuchsian. In particular, all congruence subgroups and their conjugates are Fuchsian.

Since any Fuchsian group Δ acts properly discontinuously on $\mathbb{H}$ and the hyperbolic measure is $\mathrm{SL}_2(\mathbb{R})$-invariant by Proposition 1.1, the hyperbolic measure induces a measure on the quotient $\Delta\backslash\mathbb{H}$ which we also denote by $d\mu$. Moreover, there exist fundamental domains $\mathcal{F}_\Delta$ for $\Delta\backslash\mathbb{H}$ which are integrable with respect to the hyperbolic measure so that

$$\int_{\Delta\backslash\mathbb{H}} f(z)\,d\mu = \int_{\mathcal{F}_\Delta} f(z)\,d\mu,$$

for any Δ-invariant complex function f on $\mathbb{H}$. In particular, we define the **covolume** V_Δ of Δ to be

$$V_\Delta = \int_{\Delta\backslash\mathbb{H}} d\mu.$$

A Fuchsian group Δ is said to be of the **first kind** if its covolume is finite. In this case, the integral of any Δ-invariant function f over $\Delta\backslash\mathbb{H}$ is absolutely convergent whenever f is bounded.

It turns out that the modular group is Fuchsian of the first kind, but to show this we need to produce a fundamental domain for $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$ which are integrable with respect to the hyperbolic measure. In fact, we will show that a fundamental domain is

$$\mathcal{F} = \left\{ z \in \mathbb{H} : |\mathrm{Re}(z)| \le \frac{1}{2} \text{ and } |z| \ge 1 \right\}.$$

Proposition 1.6. *$\mathcal{F}$ is a fundamental domain for $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$.*

Proof. We first show any point in $\mathbb{H}$ is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to a point in $\mathcal{F}$. So let $z \in \mathbb{H}$ and $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. Then Equation (2) gives

$$\mathrm{Im}(\gamma z) = \frac{y}{|cz+d|^2}.$$

Since $\det(\gamma) = 1$ we cannot have $c = d = 0$. As $y > 0$, it is easily checked that $|cz + d| \ge \min(y,1)$ and thus bounded away from zero. Whence there are finitely many pairs of relatively prime integers (c,d) such that $|cz+d|^2 \le M$ for any $M > 0$. So choose a γ such that it minimizes $|cz+d|$ and hence maximizes $\mathrm{Im}(\gamma z)$. In particular, $\mathrm{Im}(S\gamma z) \le \mathrm{Im}(\gamma z)$ which is to say

$$\frac{\mathrm{Im}(\gamma z)}{|\gamma z|^2} \le \mathrm{Im}(\gamma z).$$

The inequality implies $|\gamma z| \ge 1$. Since $\mathrm{Im}(T^n\gamma z) = \mathrm{Im}(\gamma z)$ for all integers n, repeating the argument above with $T^n\gamma$ in place of γ, we see that $|T^n\gamma z| \ge 1$. But T shifts the real part by 1 so we can choose n such that $|\mathrm{Re}(T^n\gamma z)| \le \frac{1}{2}$. Therefore $T^n\gamma$ maps z into $\mathcal{F}$. It remains to show that no two distinct interior points of $\mathcal{F}$ are $\mathrm{SL}_2(\mathbb{Z})$-equivalent. So suppose to the contrary that there exist distinct $z, z' \in \mathcal{F}^\circ$ and $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ such that $z' = \gamma z$. Without loss of generality, we may assume $\mathrm{Im}(z') \ge \mathrm{Im}(z)$. Then

$$\mathrm{Im}(z') = \frac{\mathrm{Im}(z)}{|cz+d|^2},$$

by Equation (2) and so $|cz+d| \leq 1$. As $|z| > 1$ and $|\mathrm{Re}(z)| < \frac{1}{2}$, short computations show that this inequality is valid only if $c = 0$. Then $\det(\gamma) = 1$ forces $a = d = \pm 1$. Finally, $|\mathrm{Re}(z)| < \frac{1}{2}$ further forces $b = 0$. But then $\gamma = \pm I$ which gives a contradiction since we assumed z and z' were distinct. $\square$

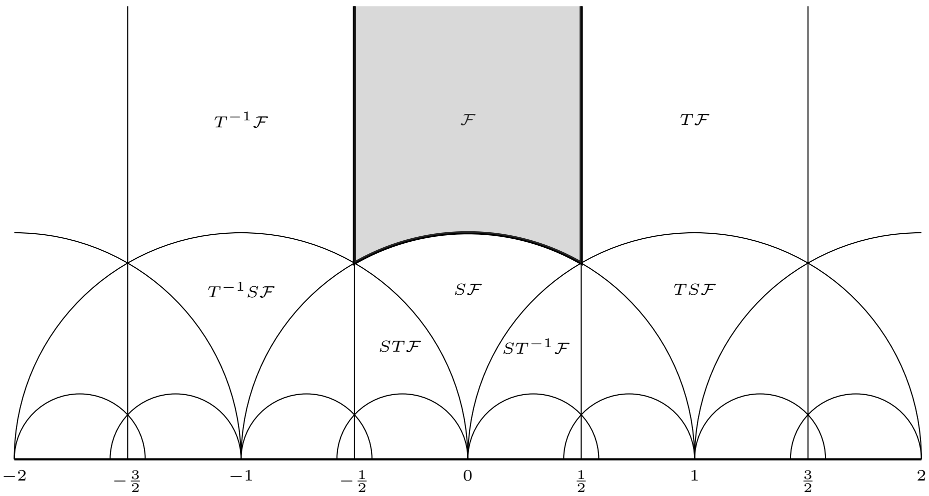

Figure 1: The standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$.

The region $\mathcal{F}$, shaded in Figure 1, is called the **standard fundamental domain**. Any $z \in \mathcal{F}$ satisfies $\mathrm{Im}(z) \geq \frac{\sqrt{3}}{2}$. Indeed, z occurs on the arc of a circle of radius $r \geq 1$ subject to $|\mathrm{Re}(z)| \leq \frac{1}{2}$. Then $\mathrm{Im}(z)$ is minimized on the arc $|z| = 1$, and on this arc

$$\mathrm{Im}(z) = \sqrt{1 - \mathrm{Re}(z)^2}.$$

The right-hand side is minimized at $\mathrm{Re}(z) = \pm\frac{1}{2}$ giving $\mathrm{Im}(z) = \frac{\sqrt{3}}{2}$. This corresponds to the points $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$ and $\frac{1}{2} + i\frac{\sqrt{3}}{2}$ which are ω and $1+\omega$ respectively. They are the vertices of the boundary of $\mathcal{F}$.

If Γ is a congruence subgroup, we can use the standard fundamental domain to produce a fundamental domain for $\Gamma\backslash\mathbb{H}$ which can be used to evaluate the covolume of Γ. It so happens that

$$\mathcal{F}_\Gamma = \bigcup_{\beta \in \Gamma\backslash\mathrm{SL}_2(\mathbb{Z})} \beta\mathcal{F},$$

is a fundamental domain for $\Gamma\backslash\mathbb{H}$ which is obtained from the standard fundamental domain using representatives of orbits of $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$.

Proposition 1.7. *$\mathcal{F}_\Gamma$ is a fundamental domain for $\Gamma\backslash\mathbb{H}$.*

Proof. We first show that every point in $\mathbb{H}$ is Γ-equivalent to a point in $\mathcal{F}_\Gamma$. From the definition of $\mathcal{F}_\Gamma$, it is straightforward to check that

$$\bigcup_{\gamma \in \Gamma} \gamma\mathcal{F}_\Gamma = \mathbb{H}.$$

Whence every point in $\mathbb{H}$ is Γ-equivalent to a point in $\mathcal{F}_\Gamma$. We will now show that no two distinct interior points of $\mathcal{F}_\Gamma$ are Γ-equivalent. Suppose to the contrary that there are distinct $z, z' \in \mathcal{F}_\Gamma^\circ$ and a $\gamma \in \Gamma$ such that $\gamma z = z'$. In particular, $z \in \beta\mathcal{F}^\circ$ and $z' \in \beta'\mathcal{F}^\circ$ for some $\beta, \beta' \in \Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$. As no two distinct interior points of $\mathcal{F}$ are $\mathrm{SL}_2(\mathbb{Z})$-equivalent, this forces β and β' to belong to the same coset of $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$. Whence $\beta = \beta'$ and implying $z, z' \in \beta\mathcal{F}^\circ \cap \beta'\mathcal{F}^\circ$. This gives a contradiction since no two distinct interior points of $\mathcal{F}$ are $\mathrm{SL}_2(\mathbb{Z})$-equivalent. Hence no two distinct interior points of $\mathcal{F}_\Gamma$ are Γ-equivalent. $\square$

We can now show that the modular group is Fuchsian of the first kind. For brevity, we write $V = V_{\mathrm{SL}_2(\mathbb{Z})}$ so that

$$V = \int_{\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}} d\mu.$$

Using Proposition 1.6 to write this integral explicitly, we obtain

$$V = \int_{-\frac{1}{2}}^{\frac{1}{2}} \int_{\sqrt{1-x^2}}^{\infty} \frac{dy\,dx}{y^2},$$

and a short computation shows $V = \frac{\pi}{3}$. Therefore the modular group is Fuchsian of the first kind. There is also a simple relation between V_Γ and d_Γ given by

$$V_\Gamma = d_\Gamma V, \tag{3}$$

which follows immediately from Proposition 1.7. Whence every congruence subgroup is Fuchsian of the first kind as well. Using $\mathrm{SL}_2(\mathbb{R})$-invariance of the hyperbolic measure and the change of variables $z \mapsto \alpha^{-1}z$, a short computation shows

$$V_{\alpha^{-1}\Gamma\alpha} = V_\Gamma, \tag{4}$$

for any $\alpha \in \mathrm{SL}_2(\mathbb{R})$. So the conjugate of every congruence subgroup is also Fuchsian of the first kind.

It is also possible to say something about the stabilizer subgroups of points in $\mathbb{H}$ with respect to the action of a congruence subgroup Γ. Let Γ_z denote the stabilizer subgroup of $z \in \mathbb{H}$. Then $\Gamma_{\gamma z} = \gamma\Gamma_z\gamma^{-1}$ for any $\gamma \in \Gamma$ and it follows that all of the stabilizer subgroups are determined by those for $z \in \mathcal{F}_\Gamma$. Moreover, since $\mathcal{F}_\Gamma$ is a fundamental domain, Γ_z is trivial if $z \in \mathcal{F}_\Gamma^\circ$. So any point with a nontrivial stabilizer subgroups must lie on the boundary of $\mathcal{F}_\Gamma$. As Γ is a subgroup of $\mathrm{SL}_2(\mathbb{Z})$, we conclude Γ_z is a subgroup of $\mathrm{SL}_2(\mathbb{Z})_z$. This leads us to determining the stabilizer subgroups in the case of the modular group.

Proposition 1.8. *Let $z \in \mathcal{F}$. Then*

$$\mathrm{SL}_2(\mathbb{Z})_z = \begin{cases} \langle S \rangle & \text{if } z = i, \\ \langle ST \rangle & \text{if } z = \omega, \\ \langle TS \rangle & \text{if } z = 1+\omega, \\ \langle -I \rangle & \text{if } z \neq i, \omega, 1+\omega. \end{cases}$$

Moreover, $\mathrm{SL}_2(\mathbb{Z})_i$ *is of order* 4 *while* $\mathrm{SL}_2(\mathbb{Z})_\omega$ *and* $\mathrm{SL}_2(\mathbb{Z})_{1+\omega}$ *are of order* 6.

Proof. Let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. Then $\gamma z = z$ implies

$$cz^2 + (d-a)z - b = 0.$$

We will analyze this identity on a case by case basis. In particular, we will take $z = i$, $z = \omega$, $z = 1+\omega$, and $z \neq i, \omega, 1+\omega$ respectively.

Taking $z = i$ in this identity gives

$$-c + (d-a)i - b = 0.$$

Equating the real and imaginary parts shows that $b = -c$ and $d = a$. Then $\gamma = \left(\begin{smallmatrix} a & -c \\ c & a \end{smallmatrix}\right)$ and $a^2 + c^2 = 1$. Therefore we have $a = 1$ and $c = 0$, $a = -1$ and $c = 0$, $a = 0$ and $c = 1$, or $a = 0$ and $c = -1$. These four cases correspond to the matrices I, $-I$, S, and $-S$ respectively. Since $S^2 = -I$, it follows that $\mathrm{SL}_2(\mathbb{Z})_i = \langle S \rangle$ and is of order 4.

Taking $z = \omega$ in the identity instead gives

$$c\omega^2 + (d-a)\omega - b = 0.$$

Recall that $m_\omega(x) = x^2 + x + 1$ is the minimal polynomial for ω over $\mathbb{Q}$. So $\omega^2 = -\omega - 1$ and substituting this identity into the one above yields

$$(d - a - c)\omega - (b + c) = 0.$$

By minimality of $m_\omega(x)$, ω and 1 are linearly independent over $\mathbb{Q}$ so we may equate coefficients to see that $d = a + c$ and $b = -c$. Then $\gamma = \left(\begin{smallmatrix} a & -c \\ c & a+c \end{smallmatrix}\right)$ and $a^2 + ac + c^2 = 1$. Completing the square shows $(a+c)^2 - ac = 1$. Therefore we have $a = 1$ and $c = 0$, $a = -1$ and $c = 0$, $a = 0$ and $c = 1$, $a = 0$ and $c = -1$, $a = -1$ and $c = 1$, or $a = 1$ and $c = -1$. These six cases correspond to the matrices I, $-I$, ST, $(ST)^4$, $(ST)^2$, and $(ST)^5$ respectively. As $(ST)^3 = -I$, it follows that $\mathrm{SL}_2(\mathbb{Z})_\omega = \langle ST \rangle$ and is of order 6. Moreover, if $z = 1 + \omega$ then $z = T\omega$. Whence $\mathrm{SL}_2(\mathbb{Z})_{1+\omega} = T\mathrm{SL}_2(\mathbb{Z})_\omega T^{-1} = \langle TS \rangle$ and is of order 6.

It remains to show that if $z \neq i, \omega, 1+\omega$ then $\mathrm{SL}_2(\mathbb{Z})_z = \langle I \rangle$. Equivalently, will show the contrapositive which is to say that if $\mathrm{SL}_2(\mathbb{Z})_z \neq \langle -I \rangle$ then $z = i, \omega, 1+\omega$. Since the stabilizer subgroup of z is trivial if $z \in \mathcal{F}^\circ$, we may additionally assume z lies on the boundary of $\mathcal{F}$. Now suppose for such z that $\gamma \neq \pm I$. We first claim $c > 0$. For if $c = 0$, $\det(\gamma) = 1$ implies $a = d = \pm 1$. This further forces $b = 0$ implying $\gamma = \pm I$ which is a contradiction. Hence $c \neq 0$ and so z satisfies the aforementioned identity which is a quadratic. A short computation shows that the discriminant is $(a+d)^2 - 4$. Since this quadratic has integral coefficients and z is complex, the discriminant must be negative. So

$$(a+d)^2 - 4 < 0,$$

which forces $a + d = 0, 1, -1$. These three cases correspond to the discriminant being -4 in the first case and -3 in the latter two cases. Whence the quadratic formula implies

$$z = \frac{a \pm i}{c}, \quad \text{or} \quad z = \frac{2a - 1 \pm \sqrt{3}i}{2c}, \quad \text{or} \quad z = \frac{2a + 1 \pm \sqrt{3}i}{2c},$$

respectively. In any case, recalling that $\mathrm{Im}(z) \geq \frac{\sqrt{3}}{2}$ for $z \in \mathcal{F}$ implies $c = \pm 1$. This forces $a = 0$ in the first case, $a = 0, 1$ in the second, and $a = 0, -1$ in the third. Whence $z = i$ in the first case and $z = \omega, 1+\omega$ in the second and third cases. This completes the proof. $\square$

Returning to the case of a congruence subgroup Γ, it follows from this result and our previous remarks that Γ_z is necessarily trivial if z is not $\mathrm{SL}_2(\mathbb{Z})$-equivalent to i, ω, or $1+\omega$. Moreover, if z is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to one of these points then Γ_z is trivial unless at least one of the matrices $\pm S$, $\pm ST$, or $\pm TS$ belongs to Γ respectively. In particular, Γ_z is always finite.

A **modular curve** is a quotient $\Gamma\backslash\mathbb{H}$ of $\mathbb{H}$ by a congruence subgroup Γ. The **level** of $\Gamma\backslash\mathbb{H}$ is the level of Γ. Observe that $\Gamma\backslash\mathbb{H}$ is not compact as the fundamental domain $\mathcal{F}_\Gamma$ need not contain points of $\widehat{\mathbb{Q}}$. These points correspond to cusps which we now define. Recall that Γ acts on $\widehat{\mathbb{Q}}$. A **cusp class** of $\Gamma\backslash\mathbb{H}$ is a coset of $\Gamma\backslash\widehat{\mathbb{Q}}$. There is only one cusp class for the modular group since it acts transitively. Indeed, $\widehat{\mathbb{Q}}$ is exactly the orbit of ∞. For if $\frac{a}{c} \in \mathbb{Q}$ with $(a,c)=1$, Bézout's identity provides the existence of integers b and d such that $ad-bc=1$ whence $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ satisfies $\gamma\infty = \frac{a}{c}$. Therefore the action is transitive and there can only be finitely many cosets of $\Gamma\backslash\widehat{\mathbb{Q}}$ since Γ has finite index in the modular group. In other words, the number of cusp classes is finite and at most d_Γ. In particular, the orbit of ∞ is a cusp class of $\Gamma\backslash\mathbb{H}$. A **cusp** is a representative of a cusp class. We denote cusps by gothic characters $\mathfrak{a}$, $\mathfrak{b}$, etc.

Remark 1.9. The cusps are exactly the points needed to make a fundamental domain $\mathcal{F}_\Gamma$ compact as a subset of $\widehat{\mathbb{C}}$. To see this, suppose $\mathfrak{a}$ is a limit point of $\mathcal{F}_\Gamma$ that does not belong to $\mathcal{F}_\Gamma$. Then $\mathfrak{a} \in \widehat{\mathbb{R}}$. As the unique limit point of $\mathcal{F}$ not belonging to $\mathcal{F}$ is ∞ and $\widehat{\mathbb{Q}} = \mathrm{SL}_2(\mathbb{Z})\infty$, Proposition 1.7 implies $\mathfrak{a} \in \widehat{\mathbb{Q}}$ which means $\mathfrak{a}$ is a cusp. In the case of the standard fundamental domain $\mathcal{F}$, the only cusp is ∞.

We will now describe the stabilizer subgroups of Γ relative to a cusp $\mathfrak{a}$. This will slightly depend upon if Γ contains $-I$, so let $\varepsilon_\Gamma = 1, 0$ according to if $-I \in \Gamma$ or not. We will denote the stabilizer subgroup of the cusp $\mathfrak{a}$ by $\Gamma_\mathfrak{a}$. We begin by describing Γ_∞. If $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma$ stabilizes ∞ then necessarily $c=0$ and since $\det(\gamma)=1$ we must have $a=d=\pm 1$ or $a=d=1$ according to if $-I \in \Gamma$ or not. In either case, γ acts by translation by b. If we let h_∞ be the smallest positive integer such that $\gamma_\infty = \left(\begin{smallmatrix} 1 & h_\infty \\ 0 & 1 \end{smallmatrix}\right) \in \Gamma_\infty$ then

$$\Gamma_\infty = \langle \gamma_\infty, (-I)^{\varepsilon_\Gamma} \rangle.$$

By our previous discussion, there exists a matrix $\eta_\mathfrak{a} \in \mathrm{SL}_2(\mathbb{Z})$ such that $\eta_\mathfrak{a}\infty = \mathfrak{a}$. This matrix is determined up to right multiplication by elements of Γ_∞. In any case, $\Gamma_\mathfrak{a} = \eta_\mathfrak{a}\Gamma_\infty\eta_\mathfrak{a}^{-1} \cap \Gamma$ and so

$$\Gamma_\mathfrak{a} = \langle \eta_\mathfrak{a}, (-I)^{\varepsilon_\Gamma} \rangle,$$

where $\gamma_\mathfrak{a}$ is of the form

$$\gamma_\mathfrak{a} = \eta_\mathfrak{a} \begin{pmatrix} 1 & h_\mathfrak{a} \\ 0 & 1 \end{pmatrix} \eta_\mathfrak{a}^{-1},$$

for some positive integer $h_\mathfrak{a}$. We call $h_\mathfrak{a}$ the **width** of the cusp $\mathfrak{a}$. Equivalently, $h_\mathfrak{a}$ is the shift of the minimal translation belonging to $\eta_\mathfrak{a}^{-1}\Gamma\eta_\mathfrak{a}$. Also observe that $h_\mathfrak{a}$ does not depend on the choice of $\eta_\mathfrak{a}$. We will set

$$\gamma_\infty^{(h_\mathfrak{a})} = \begin{pmatrix} 1 & h_\mathfrak{a} \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \Gamma_\infty^{(h_\mathfrak{a})} = \langle \gamma_\infty^{(h_\mathfrak{a})}, (-I)^{\varepsilon_\Gamma} \rangle,$$

so that

$$\eta_{\mathfrak{a}}^{-1}\Gamma_{\mathfrak{a}}\eta_{\mathfrak{a}} = \Gamma_\infty^{(h_{\mathfrak{a}})}.$$

In particular, $\gamma_\infty^{(h_\infty)} = \gamma_\infty$ and $\Gamma_\infty^{(h_\infty)} = \Gamma_\infty$.

Remark 1.10. Suppose $\mathfrak{a}'$ is a cusp that belongs to the same cusp class at $\mathfrak{a}$. Then $\mathfrak{a}' = \gamma\mathfrak{a}$ for some $\gamma \in \Gamma$. It follows that $\Gamma\eta_{\mathfrak{a}}\Gamma_\infty$ only depends upon the cusp class of $\mathfrak{a}$ which is $\Gamma\eta_{\mathfrak{a}}\infty$. Moreover, $\Gamma_{\mathfrak{a}'} = \gamma\Gamma_{\mathfrak{a}}\gamma^{-1}$ and $\Gamma_{\mathfrak{a}'} = \gamma\eta_{\mathfrak{a}}\Gamma_\infty\eta_{\mathfrak{a}}^{-1}\gamma^{-1} \cap \Gamma$. Whence $\gamma_{\mathfrak{a}'} = \gamma\gamma_{\mathfrak{a}}\gamma^{-1}$ and therefore $h_{\mathfrak{a}} = h_{\mathfrak{a}'}$ because the width of a cusp does not depend upon the choice of $\eta_{\mathfrak{a}}$. This means that the width of $\mathfrak{a}$ only depends upon its cusp class and so the same is true for $\gamma_\infty^{(h_{\mathfrak{a}})}$ and $\Gamma_\infty^{(h_{\mathfrak{a}})}$.

If Γ is of level N then N is the smallest positive integer such that $\Gamma(N) \subseteq \Gamma$. This means $\left(\begin{smallmatrix} 1 & N \\ 0 & 1 \end{smallmatrix}\right)$ is the minimal translation guaranteed to belong to Γ. Since principal congruence subgroups are normal in the modular group, we have $\Gamma(N) \subseteq \Gamma_\infty^{(h_{\mathfrak{a}})}$ and so the aforementioned translation must be a power of $\gamma_\infty^{(h_{\mathfrak{a}})}$ which means $h_{\mathfrak{a}} \mid N$. We say that Γ is **reduced** at $\mathfrak{a}$ if $h_{\mathfrak{a}} = 1$. For example, the congruence subgroups $\Gamma_0(N)$, $\Gamma_1(N)$, and $\Gamma(N)$ are all reduced at ∞. The width of the cusps are also related to the index d_Γ in a simple manner.

Proposition 1.11. *For any congruence subgroup* Γ *and any cusp* $\mathfrak{a}$ *of* $\Gamma\backslash\mathbb{H}$*, there are* $h_{\mathfrak{a}}$ *many cosets of* $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$*. In particular,*

$$\sum_{\mathfrak{a}\in\Gamma\backslash\widehat{\mathbb{Q}}} h_{\mathfrak{a}} = d_\Gamma.$$

Proof. Choose $\eta_{\mathfrak{a}} \in \mathrm{SL}_2(\mathbb{Z})$ with $\eta_{\mathfrak{a}}\infty = \mathfrak{a}$. It follows from Remark 1.10 that the cusp classes of $\Gamma\backslash\mathbb{H}$ are in bijection with the double cosets of $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})/\Gamma_\infty$ where a cusp class $\Gamma\eta_{\mathfrak{a}}\infty$ corresponds to the double coset $\Gamma\eta_{\mathfrak{a}}\Gamma_\infty$ under this bijection. As the action of $\mathrm{SL}_2(\mathbb{Z})$ on $\widehat{\mathbb{Q}}$ is transitive, consider the surjection

$$\phi : \Gamma\backslash\mathrm{SL}_2(\mathbb{Z}) \to \Gamma\backslash\mathrm{SL}_2(\mathbb{Z})/\Gamma_\infty \qquad \Gamma\eta_{\mathfrak{a}} \mapsto \Gamma\eta_{\mathfrak{a}}\Gamma_\infty.$$

We will compute the number of cosets in a fiber of ϕ. Suppose $\mathfrak{a}$ and $\mathfrak{a}'$ are Γ-equivalent cusps so that $\Gamma\eta_{\mathfrak{a}}$ and $\Gamma\eta_{\mathfrak{a}'}$ belong to the fiber of $\Gamma\eta_{\mathfrak{a}}\Gamma_\infty = \Gamma\eta_{\mathfrak{a}'}\Gamma_\infty$. Then $\eta_{\mathfrak{a}'} = \gamma\eta_{\mathfrak{a}}\gamma_\infty^n$ for some $\gamma \in \Gamma$ and integer n. Whence $\Gamma\eta_{\mathfrak{a}} = \Gamma\eta_{\mathfrak{a}'}$ if and only if $\Gamma\eta_{\mathfrak{a}} = \Gamma\eta_{\mathfrak{a}}\gamma_\infty^n$ which is to say that $\gamma_\infty^n \in \eta_{\mathfrak{a}}^{-1}\Gamma\eta_{\mathfrak{a}}$. But then $\gamma_\infty^n \in \Gamma_\infty^{(h_{\mathfrak{a}})}$ which happens if and only if $h_{\mathfrak{a}} \mid n$. So the cosets $\Gamma\eta_{\mathfrak{a}}$ and $\Gamma\eta_{\mathfrak{a}'}$ are equal if and only if $h_{\mathfrak{a}} \mid n$. This means there are $h_{\mathfrak{a}}$ cosets in the fiber of $\Gamma\eta_{\mathfrak{a}}\Gamma_\infty$. In other words, there are $h_{\mathfrak{a}}$ many cosets of $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$ proving the first statement. For the second, the sum of the number of cosets in the fibers of ϕ is equal to the order of $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$ since ϕ is surjective. This is to say

$$\sum_{\mathfrak{a}\in\Gamma\backslash\widehat{\mathbb{Q}}} h_{\mathfrak{a}} = d_\Gamma,$$

proving the second statement. ∎

As it turns out, it will be more useful deal with the width of cusps implicitly. Set

$$B_\Gamma = \langle T, (-1)^{\varepsilon_\Gamma} \rangle.$$

We call $\sigma_{\mathfrak{a}} \in \mathrm{SL}_2(\mathbb{R})$ a **scaling matrix** for $\mathfrak{a}$ if $\sigma_{\mathfrak{a}}\infty = \mathfrak{a}$ and $\sigma_{\mathfrak{a}}^{-1}\gamma_{\mathfrak{a}}\sigma_{\mathfrak{a}} = T$. This forces

$$\sigma_{\mathfrak{a}}^{-1}\Gamma_{\mathfrak{a}}\sigma_{\mathfrak{a}} = B_\Gamma.$$

The second condition is possible only because we are allowing the entries to be real. For example, a short computation shows that we can choose the scaling matrix to be

$$\sigma_{\mathfrak{a}} = \eta_{\mathfrak{a}} \begin{pmatrix} \sqrt{h_{\mathfrak{a}}} & 0 \\ 0 & \frac{1}{\sqrt{h_{\mathfrak{a}}}} \end{pmatrix}.$$

Observe that scaling matrices are determined up to right multiplication by powers of T. Also, if Γ is reduced at $\mathfrak{a}$ then we may take $\sigma_{\mathfrak{a}} = \eta_{\mathfrak{a}}$ so that the scaling matrix can be chosen to lie in $\mathrm{SL}_2(\mathbb{Z})$. In this case, $\Gamma_{\mathfrak{a}} = B_\Gamma$. For example, this occurs at ∞ for the Hecke congruence subgroups.

Lastly, let us discuss a useful property of integrals over $\Gamma\backslash\mathbb{H}$. Recall that we may use the fundamental domain $\mathcal{F}_\Gamma$ to integrate over $\Gamma\backslash\mathbb{H}$ by writing

$$\int_{\Gamma\backslash\mathbb{H}} f(z)\,d\mu = \int_{\mathcal{F}_\Gamma} f(z)\,d\mu,$$

for any Γ-invariant complex function f on $\mathbb{H}$. As $\Gamma_{\mathfrak{a}}$ is also Fuchsian of the first kind, $\Gamma_{\mathfrak{a}}\backslash\mathbb{H}$ admits fundamental domains $\mathcal{F}_{\Gamma_{\mathfrak{a}}}$ that are integrable with respect to the hyperbolic measure. So

$$\int_{\Gamma_{\mathfrak{a}}\backslash\mathbb{H}} f(z)\,d\mu = \int_{\mathcal{F}_{\mathfrak{a}}} f(z)\,d\mu,$$

for any $\Gamma_{\mathfrak{a}}$-invariant complex function f on $\mathbb{H}$. It turns out that these two types of integrals are related for complex functions that are averaged over $\Gamma_{\mathfrak{a}}\backslash\Gamma$. This is known as the **unfolding/folding method**.

Theorem (Unfolding/folding method). *Let $\mathfrak{a}$ be a cusp of $\Gamma\backslash\mathbb{H}$. If f is a $\Gamma_{\mathfrak{a}}$-invariant complex function on $\mathbb{H}$ then*

$$\int_{\Gamma\backslash\mathbb{H}} \sum_{\gamma\in\Gamma_{\mathfrak{a}}\backslash\Gamma} f(\gamma z)\,d\mu = \int_{\Gamma_{\mathfrak{a}}\backslash\mathbb{H}} f(z)\,d\mu,$$

provided either side is absolutely convergent.

Proof. If the left-hand side converges absolutely then we may interchange the sum and integral. Upon making the change of variables $z \mapsto \gamma^{-1}z$, the Γ-invariance of $d\mu$ implies that the integral takes the form

$$\sum_{\gamma\in\Gamma_{\mathfrak{a}}\backslash\Gamma} \int_{\gamma\mathcal{F}_\Gamma} f(z)\,d\mu.$$

The result follows as $\mathbb{H} = \bigcup_{\gamma\in\Gamma} \gamma\mathcal{F}$. Moreover, we can argue in reverse provided the right-hand side converges absolutely. This completes the proof. $\square$

The process of transforming the left-hand side into the right-hand side is called **unfolding** while transforming the right-hand into the left-hand side is called **folding**. In the special case of $\Gamma_\infty\backslash\mathbb{H}$, the latter integral can be computed explicitly. Indeed,

$$\mathcal{F}_{\Gamma_\infty} = \{z \in \mathbb{H} : 0 \leq \mathrm{Re}(z) \leq h_\infty \text{ and } \mathrm{Im}(z) > 0\},$$

is clearly a fundamental domain for $\Gamma_\infty\backslash\mathbb{H}$ so that

$$\int_{\Gamma_\infty\backslash\mathbb{H}} f(z)\,d\mu = \int_0^\infty \int_0^{h_\infty} f(z)\,\frac{dx\,dy}{y^2},$$

for any Γ_∞-invariant complex function f on $\mathbb{H}$.

1.4 The Bruhat Decomposition and Exponential Sums

Let $\mathfrak{a}$ and $\mathfrak{b}$ be cusps of $\Gamma\backslash\mathbb{H}$ with respective scaling matrices $\sigma_\mathfrak{a}$ and $\sigma_\mathfrak{b}$ respectively. It will be essential to have a double coset decomposition for sets of the form $\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$. This is known as the **Bruhat decomposition** for $\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$.

Theorem (Bruhat decomposition). *Let $\mathfrak{a}$ and $\mathfrak{b}$ be cusps of $\Gamma\backslash\mathbb{H}$ with scaling matrices $\sigma_\mathfrak{a}$ and $\sigma_\mathfrak{b}$ respectively. Then we have the disjoint decomposition*

$$\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b} = \Omega_\infty \cup \left(\bigcup_{\substack{c \in \mathbb{Z}_+ \\ d \pmod{c}}} \Omega_{c/d} \right),$$

where

$$\Omega_\infty = B_\Gamma w_\infty B_\Gamma \quad \text{and} \quad \Omega_{c/d} = B_\Gamma \omega_{c/d} B_\Gamma,$$

with $\omega_\infty = \left(\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right) \in \sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$ and $\omega_{c/d} = \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right) \in \sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$ if such matrices exists otherwise Ω_∞ and $\Omega_{c/d}$ are empty respectively. In the former case for Ω_∞, ω_∞ exists if and only if $\mathfrak{a}$ and $\mathfrak{b}$ are Γ-equivalent. In the former case for $\Omega_{c/d}$, a and d are determined modulo c and $\Omega_{c/d}$ does not depend on the entry a of $\omega_{c/d}$.

Proof. As $B_\Gamma = \sigma_\mathfrak{a}^{-1}\Gamma_\mathfrak{a}\sigma_\mathfrak{a}$ and $B_\Gamma = \sigma_\mathfrak{b}^{-1}\Gamma_\mathfrak{b}\sigma_\mathfrak{b}$, short computations show that $\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$ is invariant under left and right multiplication by elements B_Γ. This proves Ω_∞ and $\Omega_{c/d}$ are subsets of $\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$. Now any $\omega_\infty \in \sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$ satisfies $\omega_\infty = \sigma_\mathfrak{a}^{-1}\gamma\sigma_\mathfrak{b}$ for some $\gamma \in \Gamma$. As ω_∞ fixes ∞, a short computation shows

$$\gamma\mathfrak{b} = \mathfrak{a}.$$

This means $\mathfrak{a}$ and $\mathfrak{b}$ are Γ-equivalent. Conversely, if $\mathfrak{a}$ and $\mathfrak{b}$ are Γ-equivalent then $\gamma\mathfrak{b} = \mathfrak{a}$ for some $\gamma \in \Gamma$ and we may take $\omega_\infty = \sigma_\mathfrak{a}^{-1}\gamma\sigma_\mathfrak{b}$ because this matrix fixes ∞. This proves that ω_∞ exists if and only if $\mathfrak{a}$ and $\mathfrak{b}$ are Γ-equivalent. In this case, we may write

$$\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b} = \sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{a}\omega_\infty \quad \text{and} \quad \sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b} = \omega_\infty\sigma_\mathfrak{b}^{-1}\Gamma\sigma_\mathfrak{b},$$

from which we see that the matrices in $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}$ acting by translation are exactly Ω_∞. Every other matrix in $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}$ belongs to one of the double cosets $\Omega_{c/d}$. The relation

$$T\omega_{c/d}T = \begin{pmatrix} a+c & * \\ c & d+c \end{pmatrix},$$

shows that $\Omega_{c/d}$ is generated uniquely by a matrix of the form $\omega_{c/d} = \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right) \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}$ where a and d are determined modulo c. To see that $\Omega_{c/d}$ does not depend on the entry a of $\omega_{c/d}$, let $\omega'_{c/d} = \left(\begin{smallmatrix} a' & * \\ c & d \end{smallmatrix}\right) \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}$. Write $\omega_{c/d} = \sigma_{\mathfrak{a}}^{-1}\gamma\sigma_{\mathfrak{b}}$ and $\omega'_{c/d} = \sigma_{\mathfrak{a}}^{-1}\gamma'\sigma_{\mathfrak{b}}$ for some $\gamma, \gamma' \in \Gamma$. The relations

$$\omega_{c/d}(\omega'_{c/d})^{-1} = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \quad \text{and} \quad \gamma(\gamma')^{-1} = \sigma_{\mathfrak{a}}\omega_{c/d}(\omega'_{c/d})^{-1}\sigma_{\mathfrak{a}}^{-1},$$

imply $\gamma(\gamma')^{-1} \in \Gamma_{\mathfrak{a}}$. So $\omega_{c/d}(\omega'_{c/d})^{-1} \in B_\Gamma$ whence $B_\Gamma\omega_{c/d}B_\Gamma = B_\Gamma\omega'_{c/d}B_\Gamma$. $\square$

As the double cosets Ω_∞ and $\Omega_{c/d}$ in the Bruhat decomposition may very well be empty, we would like to track exactly which are not. To do this, let $\mathcal{C}_{\mathfrak{a},\mathfrak{b}}$ and $\mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)$ be the sets

$$\mathcal{C}_{\mathfrak{a},\mathfrak{b}} = \left\{c \in \mathbb{Z}_+ : \begin{pmatrix} * & * \\ c & * \end{pmatrix} \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}\right\} \quad \text{and} \quad \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c) = \left\{d \in \mathbb{Z}/c\mathbb{Z} : \begin{pmatrix} * & * \\ c & d \end{pmatrix} \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}\right\},$$

where $\mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)$ is defined for $c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}$. These sets are constructed so that $\mathcal{C}_{\mathfrak{a},\mathfrak{b}}$ and $\mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)$ run over precisely the pairs (c, d) for which $\Omega_{c/d}$ is nonempty. Indeed, we may write

$$\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}} = \Omega_\infty \cup \left(\bigcup_{\substack{c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}} \\ d \in \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)}} \Omega_{c/d}\right),$$

where none of the double cosets $\Omega_{c/d}$ are empty.

We will now introduce two type of exponential sum associated to pairs of cusps. These are types of Kloosterman and Salié sums. We being with the Kloosterman sums. For any $c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}$ and integers m and n, the **Kloosterman sum** $K_{\mathfrak{a},\mathfrak{b}}(m, n, c)$ relative to $\mathfrak{a}$ and $\mathfrak{b}$ is defined by

$$K_{\mathfrak{a},\mathfrak{b}}(m, n, c) = \sum_{\substack{d \in \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c) \\ \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right) \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}}} e\left(\frac{ma + nd}{c}\right).$$

This sum is well-defined since a and d are determined modulo c by the Bruhat decomposition. If $m = n = 0$ then

$$K_{\mathfrak{a},\mathfrak{b}}(0, 0, c) = |\mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)|.$$

The Salié sums are defined in a similar manner. For any $c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}$, integers m and n, and Dirichlet character χ modulo c, the **Salié sum** $S_{\chi,\mathfrak{a},\mathfrak{b}}(m, n, c)$ relative to $\mathfrak{a}$ and $\mathfrak{b}$ is defined by

$$S_{\chi,\mathfrak{a},\mathfrak{b}}(m, n, c) = \sum_{\substack{d \in \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c) \\ \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right) \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}}} \chi(d)e\left(\frac{ma + nd}{c}\right).$$

Similar as before, this sum is well-defined since a and d are determined modulo c by the Bruhat decomposition and that the χ is a character modulo c.

Hecke Congruence Subgroups

In the case of the Hecke congruence subgroup $\Gamma_0(N)$, the Bruhat decomposition simplifies and we recover the usual Kloosterman and Salié sums. As the entries a and d in $\omega_{c/d} = \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right)$ are determined modulo c, we find that d is the inverse of a modulo c because $\det(\omega_{c/d}) = 1$. Also, $\mathcal{C}_{\infty,\infty} = N\mathbb{Z}_+$ and $\mathcal{D}_{\infty,\infty}(c) = (\mathbb{Z}/c\mathbb{Z})^*$ because if $\gamma = \left(\begin{smallmatrix} * & * \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$ then $c \equiv 0 \pmod N$ and we must have $(c, d) = 1$ as $\det(\gamma) = 1$. Moreover, every such pair gives rise to a matrix in $\Gamma_0(N)$. Whence

$$K_{\infty,\infty}(m, n, c) = K(m, n, c) \quad \text{and} \quad S_{\chi,\infty,\infty}(m, n, c) = S_\chi(m, n, c),$$

which are the usual Kloosterman and Salié sums respectively. In these cases, we will suppress the dependencies upon the cusps.

2 Modular Forms

If a is an integer modulo m with $(a, m) = 1$ we write a^{-1} for the inverse of a modulo m. For any complex z let

$$e(z) = e^{2\pi i z}.$$

We will distinguish the primitive third root of unity $\omega = e\left(\frac{1}{3}\right)$. Also, all sums and products over a quotient group will be consider to be over a complete set of representatives.

2.1 Modular Forms

For all all $\alpha = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{R})$ and $z \in \mathbb{H}$, define $j(\alpha, z)$ by

$$j(\alpha, z) = (cz + d).$$

With this notation, Equation (1) becomes

$$\mathrm{Im}(\alpha z) = \det(\alpha)\frac{\mathrm{Im}(z)}{|j(\alpha, z)|^2}. \tag{5}$$

There is a very useful property that $j(\alpha, z)$ satisfies. To state it, let $\alpha' = \left(\begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{R})$. Then

$$\alpha\alpha' = \begin{pmatrix} a'a + b'c & a'b + b'b \\ c'a + d'c & c'b + d'd \end{pmatrix},$$

and from this identity, a short computation shows

$$j(\alpha'\alpha, z) = j(\alpha', \alpha z)j(\alpha, z).$$

This identity is called the **cocycle condition** for $j(\alpha, z)$. For any nonnegative integer k and $\alpha \in \mathrm{GL}_2^+(\mathbb{R})$, we define the **slash operator** $|_k\alpha : C(\mathbb{H}) \to C(\mathbb{H})$ to be the linear operator given by

$$(f|_k\alpha)(z) = \det(\alpha)^{\frac{k}{2}} j(\alpha, z)^{-k} f(\alpha z).$$

In the case of $\alpha = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{R})$, Equation (1) simplifies to

$$\operatorname{Im}(\alpha z) = \frac{\operatorname{Im}(z)}{|j(\alpha, z)|^2}. \tag{6}$$

The slash operator also takes the simplified form

$$(f|_k\alpha)(z) = j(\alpha, z)^{-k} f(\alpha z).$$

In any case, the cocycle condition implies that the slash operator is multiplicative as short computations show

$$(f|_k\alpha'\alpha)(z) = ((f|_k\alpha')|_k\alpha)(z) \quad \text{and} \quad ((fg)|_{k+\ell}\alpha)(z) = (f|_k\alpha)(z)(g|_\ell\alpha)(z),$$

for $f, g \in C(\mathbb{H})$, nonnegative integers k and ℓ, and any $\alpha, \alpha' \in \mathrm{GL}_2^+(\mathbb{Q})$. For fixed k, an operator on $C(\mathbb{H})$ is said to be **invariant** if it commutes with the slash operator $|_k\alpha$ for every $\alpha \in \mathrm{SL}_2(\mathbb{Q})$.

We can now introduce modular forms. Let Γ be a congruence subgroup of level N and $\vartheta : \Gamma \to \mathbb{C}^*$ be a finite order homomorphism. We say that a function $f : \mathbb{H} \to \mathbb{C}$ is **modular form** on $\Gamma\backslash\mathbb{H}$ of **weight** k, **level** N, and **nebentypus** ϑ if the following properties are satisfied:

(i) f is holomorphic on $\mathbb{H}$.

(ii) $(f|_k\gamma)(z) = \vartheta(\gamma)f(z)$ for all $\gamma \in \Gamma$.

(iii) $(f|_k\gamma)(z) = O(1)$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$.

Furthermore, we say f is a **cusp form** if the additional property is satisfied:

(iv) For all cusps $\mathfrak{a}$ and positive y, we have

$$\int_0^1 (f|_k\sigma_{\mathfrak{a}})(z)\, dx = 0.$$

Property (ii) is called the **modularity condition** and we say f is **modular**. The modularity condition can equivalently be expressed as

$$f(\gamma z) = \vartheta(\gamma)j(\gamma, z)^k f(z).$$

We write n_ϑ for the order of the nebentypus so that ϑ takes its values in the n_ϑ-th roots of unity. Moreover, we say that ϑ is **regular** at a cusp $\mathfrak{a}$ if $\vartheta(\gamma_{\mathfrak{a}}) = 1$. Equivalently, $\vartheta(\gamma) = 1$ for every $\gamma \in \Gamma_{\mathfrak{a}}$. Property (iii) is called the **growth condition** and we say f is **holomorphic at the cusps**. By modularity, it suffices to verify the growth condition for a set of scaling matrices associated to a complete set of Γ-inequivalent cusps. Therefore we only have to check this condition for finitely many matrices since the number of cusp classes is finite. Also, the growth condition is equivalent to $(f|_k\alpha)(z) = O(1)$ for all $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. To see this, we require a lemma.

Lemma 2.1. *Every $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ can be written as $\alpha = \gamma\eta$ for some $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and $\eta \in \mathrm{GL}_2^+(\mathbb{Q})$ with $\eta = \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right)$.*

Proof. Let r be a positive integer such that $r\alpha = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{Q})$ has integral entries. As $r\alpha$ has nonzero determinant, we cannot have $a = c = 0$. Whence we may set $a' = \frac{a}{(a,c)}$ and $c' = \frac{c}{(a,c)}$ so that a' and c' are relatively prime integers. Then there exists $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ with $\gamma^{-1} = \left(\begin{smallmatrix} * & * \\ -c' & a' \end{smallmatrix}\right)$. Moreover, a short computation shows

$$\gamma^{-1} r\alpha = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}.$$

The claim is complete upon setting $\eta = \gamma^{-1}\alpha$. $\square$

For the equivalence, use this lemma to write $\alpha = \gamma\eta$ for some $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and $\eta \in \mathrm{GL}_2^+(\mathbb{Q})$ with $\eta = \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right)$. The cocycle condition gives

$$(f|_k\alpha)(z) = \det(\eta)^{\frac{k}{2}} j(\eta, z)(f|_k\gamma)(\eta z),$$

and as $j(\eta, z)$ is a constant, we have $(f|_k\alpha)(z) = O(1)$ proving the forward implication. The reverse implication is clear since $\mathrm{SL}_2(\mathbb{Z}) \subset \mathrm{GL}_2^+(\mathbb{Q})$.

Remark 2.2. If f is a weight k modular form with nebentypus ϑ on $\Gamma\backslash\mathbb{H}$ and $-I \in \Gamma$ then modularity implies

$$f(z) = \vartheta(-I)(-1)^k f(z).$$

Therefore $\vartheta(-I) = (-1)^k$. This means ϑ cannot be the identity if k is odd. In particular, there are no nonzero modular forms of odd weight with trivial nebentypus on $\mathrm{SL}_2(\mathbb{Z})$.

Modular forms also admit Fourier series expansions. First observe that modularity implies

$$(f|_k\sigma_{\mathfrak{a}})(z+1) = \vartheta(\gamma_{\mathfrak{a}})(f|_k\sigma_{\mathfrak{a}})(z),$$

which is to say $f|_k\sigma_{\mathfrak{a}}$ is 1-periodic up to the constant factor $\vartheta(\gamma_{\mathfrak{a}})$. This means we only need to verify the growth condition as $y \to \infty$. In any case, as $f|_k\sigma_{\mathfrak{a}}$ is also smooth, it admits an absolutely and uniformly convergent Fourier series expansion. So we may write

$$(f|_k\sigma_{\mathfrak{a}})(z) = \sum_{n \geq 0} a_{\mathfrak{a}}(y, n)e(nx),$$

where the sum is only over nonnegative n because holomorphy at the cusps forces $f|_k\sigma_{\mathfrak{a}}$ to be bounded. We can also simplify the Fourier coefficients $a_{\mathfrak{a}}(y, n)$. Indeed, as $f|_k\sigma_{\mathfrak{a}}$ is holomorphic it satisfies the Cauchy-Riemann equations so that

$$\frac{1}{2}\left(\frac{\partial f|_k\sigma_{\mathfrak{a}}}{\partial x} + i\frac{\partial f|_k\sigma_{\mathfrak{a}}}{\partial y}\right) = 0.$$

Substituting in the Fourier series expansion and equating coefficients we obtain the ODE

$$2\pi n a_{\mathfrak{a}}(y, n) + a'_{\mathfrak{a}}(y, n) = 0.$$

Solving this ODE by separation of variables, there exist coefficients $a_{\mathfrak{a}}(n)$ such that

$$a_{\mathfrak{a}}(y,n) = a_{\mathfrak{a}}(n)e(n(iy)).$$

The coefficients $a_{\mathfrak{a}}(n)$ are the only part of the Fourier series expansion depending on the cusp $\mathfrak{a}$ and scaling matrix $\sigma_{\mathfrak{a}}$. Using these coefficients instead, f admits a **Fourier series** expansion at the $\mathfrak{a}$ cusp given by

$$(f|_k\sigma_{\mathfrak{a}})(z) = \sum_{n\geq 0} a_{\mathfrak{a}}(n)e(nz),$$

with **Fourier coefficients** $a_{\mathfrak{a}}(n)$. In the case $\mathfrak{a} = \infty$ and $\sigma_{\mathfrak{a}} = I$, we will drop these dependences by writing $f|_k\sigma_{\mathfrak{a}} = f$ and $a_{\mathfrak{a}}(n) = a(n)$. The Fourier coefficients do not depend upon the choice of scaling matrix used for $f|_k\sigma_{\mathfrak{a}}$ because

$$(f|_k\sigma_{\mathfrak{a}}T)(z) = (f|_k\sigma_{\mathfrak{a}})(z+1). \tag{7}$$

Also, from property (iv) we find that f is a cusp form if and only if $a_{\mathfrak{a}}(0) = 0$ for every cusp $\mathfrak{a}$. Moreover, the Fourier coefficients corresponding to Γ-equivalent cusps have a simple relationship. Suppose $\mathfrak{b}$ and $\mathfrak{a}$ are Γ-equivalent cusps so that $\gamma\mathfrak{b} = \mathfrak{a}$ for some $\gamma \in \Gamma$. Then for associated scaling matrices $\sigma_{\mathfrak{a}}$ and $\sigma_{\mathfrak{b}}$, we have $\sigma_{\mathfrak{b}} = \gamma\sigma_{\mathfrak{a}}$. Whence modularity gives

$$(f|_k\sigma_{\mathfrak{b}})(z) = \vartheta(\gamma)(f|_k\sigma_{\mathfrak{a}})(z),$$

so that the n-th Fourier coefficient is multiplied by the root of unity $\vartheta(\gamma)$. In particular, the condition $a_{\mathfrak{a}} = 0$ only depends on the cusp class of $\mathfrak{a}$. So we only need to verify property (iv) on a complete set of Γ-inequivalent cusps. Of course, if $\vartheta(\gamma) = 1$ the Fourier coefficients are equal. For example, this occurs when the nebentypus is trivial.

We can also derive a bound for the Fourier coefficients of f at the $\mathfrak{a}$ cusp provided f is a cusp form. This bound is quite straightforward to deduce and turns out to almost be optimal. To this end, consider the integral

$$\int_0^1 (f|_k\sigma_{\mathfrak{a}})(z)\mathrm{Im}(z)^{\frac{k}{2}}e(-nz)\,dx.$$

On the one hand, substitute the Fourier series expansion and use absolute convergence to interchange the sum and the integral. In view of the identity

$$\int_0^1 e((n-m)x)\,dx = \delta_{n-m,0}, \tag{8}$$

for positive integers n and m, every term integrates to zero except the n-th term which is $a_{\mathfrak{a}}(n)\mathrm{Im}(z)^{\frac{k}{2}}$. On the other hand, holomorphy at ∞ implies that the integral is $O\left(e^{-2\pi n\mathrm{Im}(z)}\right)$. Whence

$$a_{\mathfrak{a}}(n)\mathrm{Im}(z)^{\frac{k}{2}} \ll e^{-2\pi n\mathrm{Im}(z)}.$$

Upon setting $\mathrm{Im}(z) = \frac{1}{n}$, the right-hand side is uniformly bounded and we obtain

$$a_{\mathfrak{a}}(n) \ll n^{\frac{k}{2}}.$$

This estimate is known as the **Hecke bound** for modular forms. It follows from the Hecke bound that

$$(f|_k\sigma_{\mathfrak{a}})(z) = O\left(e^{-2\pi y}\right),$$

upon summing the resulting geometric series. This implies $f|_k\sigma_{\mathfrak{a}}$ exhibits exponential decay and we say that f exhibits **exponential decay at the cusps**. This is stronger than simply knowing $f|_k\sigma_{\mathfrak{a}}$ is bounded on $\mathbb{H}$ and tends to zero as $y \to \infty$ which is provided by the fact that f is a cusp form.

Hecke Congruence Subgroups

In the case of the Hecke congruence subgroup $\Gamma_0(N)$, there are a few important simplifications when dealing with modular forms on $\Gamma_0(N)\backslash\mathbb{H}$. The first, which we have already noted, is that $\Gamma_0(N)$ is reduced ∞. Therefore $\Gamma_0(N)_\infty = B_{\Gamma_0(N)}$ and we may take $\sigma_\infty = I$ for the scaling matrix at ∞. The second is that the nebentypus can be induced by a Dirichlet character χ modulo N. Let q be the conductor of χ so that $q \mid N$. For any $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$ we know that $c \equiv 0 \pmod{N}$ whence a is the inverse of d modulo N and thus $(d, q) = 1$. We can define a function $\vartheta_\chi : \Gamma_0(N) \to \mathbb{C}^*$ by

$$\vartheta_\chi(\gamma) = \chi(d).$$

If $\gamma' = \left(\begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix}\right) \in \Gamma_0(N)$, then the relations

$$\gamma\gamma' \equiv \begin{pmatrix} * & * \\ 0 & d'd \end{pmatrix} \pmod{N} \quad \text{and} \quad \gamma^{-1} = \begin{pmatrix} d & * \\ 0 & a \end{pmatrix} \pmod{N},$$

show that

$$\vartheta_\chi(\gamma\gamma') = \vartheta_\chi(\gamma)\vartheta_\chi(\gamma') \quad \text{and} \quad \vartheta_\chi(\gamma^{-1}) = \vartheta_\chi(\gamma)^{-1}.$$

respectively. In other words, ϑ_χ is a homomorphism and it is necessarily of finite order because χ is. Therefore ϑ_χ satisfies the assumptions of a nebentypus. It is also regular at ∞. In the case f has nebentypus ϑ_χ, we simply say that f has **character** χ and write

$$\chi(\gamma) = \vartheta_\chi(\gamma).$$

2.2 Poincaré Series and Eisenstein Series

We will now introduce specific modular forms over $\Gamma\backslash\mathbb{H}$. These modular forms will be constructed by averaging the slash operator over $\Gamma_{\mathfrak{a}}\backslash\Gamma$ for a cusp $\mathfrak{a}$ of $\Gamma\backslash\mathbb{H}$ and are called Poincaré series and Eisenstein series where the latter is a special case of the former.

Fix an integer $k \geq 3$ and a cusp $\mathfrak{a}$ with scaling matrix $\sigma_{\mathfrak{a}}$. Let m be a nonnegative integer, $\vartheta : \Gamma \to \mathbb{C}^*$ be a finite order homomorphism that is regular at $\mathfrak{a}$. We define the m-th (modular) **Poincaré series** $P_{m,\vartheta,\mathfrak{a}}(z)$ of **index** m and weight k with nebentypus ϑ on $\Gamma\backslash\mathbb{H}$ at the $\mathfrak{a}$ cusp by

$$P_{m,\vartheta,\mathfrak{a}}(z) = \sum_{\gamma \in \Gamma_{\mathfrak{a}}\backslash\Gamma} \overline{\vartheta}(\gamma)\left(e(m\,\cdot)\Big|_k \sigma_{\mathfrak{a}}^{-1}\gamma\right)(z).$$

We call m the **index** of $P_{m,\vartheta,\mathfrak{a}}(z)$. Also, we must assume $k \geq 3$ is because for $k = 0, 1, 2$ the Poincaré series need not converge for any $z \in \mathbb{H}$.

The Poincaré series $P_{m,\vartheta,\mathfrak{a}}(z)$ is well-defined as it is independent of the choice of representatives γ and $\sigma_{\mathfrak{a}}$. For the nebentypus, this is easily checked by direct computation using the fact that ϑ is regular at $\mathfrak{a}$. For the exponential factor, the set of representatives for $\sigma_{\mathfrak{a}}^{-1}\gamma$ is $B_\Gamma\sigma_{\mathfrak{a}}^{-1}\gamma$ in view of $\Gamma_{\mathfrak{a}} = \sigma_{\mathfrak{a}}B_\Gamma\sigma_{\mathfrak{a}}^{-1}$. So it suffices to show

$$\left(e(m\,\cdot)\bigg|_k T\right)(z) = e(mz),$$

which is obvious. Whence $P_{m,\vartheta,\mathfrak{a}}(z)$ is well-defined. To see that $P_{m,\vartheta,\mathfrak{a}}(z)$ is holomorphic on $\mathbb{H}$, the Bruhat decomposition for $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_\infty$ and bounding term by term gives

$$P_{m,\vartheta,\mathfrak{a}}(z) \ll \sum_{(c,d)\in\mathbb{Z}^2_{0+}-\{\mathbf{0}\}} \frac{1}{|cz+d|^k}.$$

This latter series is locally absolutely uniformly convergent for $z \in \mathbb{H}$ as $k \geq 3$. Hence $P_{m,\vartheta,\mathfrak{a}}(z)$ is holomorphic on $\mathbb{H}$. As for modularity, let $\eta \in \Gamma$. A short computation using the change of variables $\eta \mapsto \eta\gamma^{-1}$ shows

$$(P_{m,\vartheta,\mathfrak{a}}|_k\eta)(z) = \vartheta(\eta)P_{m,\vartheta,\mathfrak{a}}(z).$$

To verify the growth condition we will need a technical lemma.

Lemma 2.3. *Let $a, b > 0$ and consider set*

$$S_{a,b} = \{z \in \mathbb{H} : |\mathrm{Re}(z)| \leq a \text{ and } \mathrm{Im}(z) \geq b\}.$$

Then there exists a $\delta \in (0,1)$ such that

$$|nz+m| \geq \delta|ni+m|,$$

for all integers n and m and all $z \in S_{a,b}$.

Proof. If $n = 0$ then the claim is obvious as any δ is sufficient. If $n \neq 0$ then it is equivalent to prove

$$\left|\frac{z+\frac{m}{n}}{i+\frac{m}{n}}\right| \geq \delta.$$

To this end, consider the function

$$f(z,r) = \left|\frac{z+r}{i+r}\right|,$$

for $z \in S_{a,b}$ and real r. We will prove the stronger statement $f(z,r) \geq \delta$. Clearly $f(z,r)$ is continuous on $S_{a,b} \times \mathbb{R}$ and it is also positive there since $z - r \neq 0$. A short computation shows

$$f(z,r)^2 \geq \frac{|z|^2+r^2}{1+r^2}.$$

Since $\frac{r^2}{1+r^2} \to 1$ as $r \to \pm\infty$, choose $Y > b$ such that $\frac{r^2}{1+r^2} \geq \frac{1}{4}$ for $|r| > Y$. Now let

$$S_{a,b}^{Y} = \{z \in \mathbb{H} : |\mathrm{Re}(z)| \leq a \text{ and } b \leq \mathrm{Im}(z) \leq Y\},$$

so that

$$f(z,r) \geq \frac{1}{2},$$

outside of $S_{a,b}^{Y} \times [-Y, Y]$. But this latter region is compact and so $f(z,r)$ attains a minimum $\delta' > 0$ on it. Setting $\delta = \min\left(\frac{1}{2}, \delta'\right)$ completes the proof. □

We can now verify the growth condition for $P_{m,\vartheta,\mathfrak{a}}(z)$. Choose a cusp $\mathfrak{b}$ and an associated scaling matrix $\sigma_{\mathfrak{b}}$. Using the cocycle condition, Bruhat decomposition for $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}$, and bounding term by term shows give

$$(P_{m,\vartheta,\mathfrak{a}}|_k \sigma_{\mathfrak{b}})(z) \ll \sum_{(c,d)\in\mathbb{Z}_{0+}^2 - \{\mathbf{0}\}} \frac{1}{|cz+d|^k}.$$

Now decompose this last sum as

$$\sum_{(c,d)\in\mathbb{Z}_{0+}^2 - \{\mathbf{0}\}} \frac{1}{|cz+d|^k} = \sum_{d\geq 1} \frac{1}{d^k} + \sum_{c\geq 1}\sum_{d\geq 0} \frac{1}{|cz+d|^k}.$$

Since the first sum on the right-hand side is bounded, it suffices to show that the double sum is too as $z \to \infty$. So let $\mathrm{Im}(z) \geq 1$ and δ be as in Lemma 2.3. Then for any positive integer N, we may write

$$\sum_{c\geq 1}\sum_{d\geq 0} \frac{1}{|cz+d|^k} \leq \sum_{c+d\leq N} \frac{1}{|cz+d|^k} + \frac{1}{\delta^k} \sum_{c+d>N} \frac{1}{|ci+d|^k}.$$

As $k \geq 3$, the second sum on the right-hand side is the tail of an absolute convergent series. Whence it tends to zero as $N \to \infty$. As for the first sum, it is finite and each term is bounded. Together this shows that the double sum is bounded as $z \to \infty$ verifying the growth condition. We collect this work in the following theorem:

Theorem 2.4. *The Poincaré series*

$$P_{m,\vartheta,\mathfrak{a}}(z) = \sum_{\gamma\in\Gamma_{\mathfrak{a}}\backslash\Gamma} \overline{\vartheta}(\gamma) \left(e(m\,\cdot) \bigg|_k \sigma_{\mathfrak{a}}^{-1}\gamma \right)(z),$$

is a weight k modular form with nebentypus ϑ on $\Gamma\backslash\mathbb{H}$.

In the case of index zero Poincaré series, we write $E_{\vartheta,\mathfrak{a}}(z) = P_{0,\vartheta,\mathfrak{a}}(z)$ and call $E_{\vartheta,\mathfrak{a}}(z)$ the (modular) **Eisenstein series** of weight k and nebentypus ϑ on $\Gamma\backslash\mathbb{H}$ at the $\mathfrak{a}$ cusp. It is defined by

$$E_{\vartheta,\mathfrak{a}}(z) = \sum_{\gamma\in\Gamma_{\mathfrak{a}}\backslash\Gamma} \overline{\vartheta}(\gamma) \left(1 \bigg|_k \sigma_{\mathfrak{a}}^{-1}\gamma \right)(z).$$

Then we obtain the following theorem as a special case of the former one:

Theorem 2.5. *The Eisenstein series*

$$E_{\vartheta,\mathfrak{a}}(z) = \sum_{\gamma\in\Gamma_{\mathfrak{a}}\backslash\Gamma} \overline{\vartheta}(\gamma) \left(1 \bigg|_k \sigma_{\mathfrak{a}}^{-1}\gamma \right)(z),$$

is a weight k modular form with nebentypus ϑ on $\Gamma\backslash\mathbb{H}$.

2.3 The Valence Formula

Let $\mathcal{M}_k(\Gamma,\vartheta)$ denote the complex vector space of all weight k modular forms with nebentypus ϑ on $\Gamma\backslash\mathbb{H}$ and write $\mathcal{S}_k(\Gamma,\vartheta)$ for the corresponding subspace of cusp forms. In either case, if ϑ is trivial we will suppress this dependence. Note that if Γ_1 and Γ_2 are two congruence subgroups such that $\Gamma_1 \subseteq \Gamma_2$ then we have the inclusions

$$\mathcal{M}_k(\Gamma_2,\vartheta) \subseteq \mathcal{M}_k(\Gamma_1,\vartheta) \quad \text{and} \quad \mathcal{S}_k(\Gamma_2,\vartheta) \subseteq \mathcal{S}_k(\Gamma_1,\vartheta).$$

The vector space $\mathcal{M}_k(\Gamma,\vartheta)$, and hence $\mathcal{S}_k(\Gamma,\vartheta)$ as well, turns out to be finite dimensional. This will be accomplished by showing that any $f \in \mathcal{M}_k(\Gamma,\vartheta)$ is determined by finitely many of its Fourier coefficients. In fact, this will follow from an exact formula relating the weight k, the index d_Γ, and the zeros of f. This is known as the valence formula and we require some notation to state it. For any $z \in \mathbb{H}$, let $v_z(f)$ be the index of the first nonzero term in the Taylor series expansion of f about z. Equivalently, $v_z(f)$ is the order of vanishing of f at z. Note that $v_z(f)$ is necessarily nonnegative as f is holomorphic and

$$v_{\gamma z}(f) = v_z(f),$$

for any $\gamma \in \Gamma$ by modularity. For a cusp $\mathfrak{a}$ of $\Gamma\backslash\mathbb{H}$, we let $v_{\mathfrak{a}}(f)$ be the index of the first nonzero term in the Fourier series expansion of f at the $\mathfrak{a}$ cusp. That is, $v_{\mathfrak{a}}(f)$ is the order of vanishing of $f|_k\sigma_{\mathfrak{a}}$ at ∞. To show $v_{\mathfrak{a}}(f)$ is well-defined, we need to verify that it is independent of the choice of scaling matrix $\sigma_{\mathfrak{a}}$. This follows from Equation (7) since scaling matrices are determined up to right multiplication by elements of Γ_∞. Also, $v_{\mathfrak{a}}(f)$ is again nonnegative by the holomorphy of f and

$$v_{\mathfrak{a}'}(f) = v_{\mathfrak{a}}(f),$$

for any cusp $\mathfrak{a}'$ in the same cusp class as $\mathfrak{a}$ by modularity.

Our method for proving the valance formula will be to prove it for the modular group via an application of the argument principle and then extend it to the congruence subgroup Γ. In order to do this, we will need to show how to construct a modular form on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$ given one on $\Gamma\backslash\mathbb{H}$. The following lemma provides the answer:

Lemma 2.6. *Let $f \in \mathcal{M}_k(\Gamma,\vartheta)$ and set*

$$F(z) = \prod_{\beta \in \Gamma\backslash \mathrm{SL}_2(\mathbb{Z})} ((f|_k\beta)(z))^{n_\vartheta}.$$

Then $F \in \mathcal{M}_{kn_\vartheta d_\Gamma}(\mathrm{SL}_2(\mathbb{Z}))$.

Proof. If f is zero then so is F and the claim is obvious. So assume f is nonzero. First we must show that F is well-defined which is to say that it is independent of the representatives β. Indeed, if β and β' belong to the same orbit then $\beta'\beta^{-1} \in \Gamma$. Modularity and that ϑ is of order n_ϑ together show

$$((f|_k\beta')(z))^{n_\vartheta} = ((f|_k\beta)(z))^{n_\vartheta},$$

which proves F is well-defined. Since the product is finite, holomorphy of F is clear. To prove modularity, let $\gamma \in \mathrm{SL}_2(\mathbb{Z})$. Because β runs over a complete set of representatives for $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$ so does $\beta\gamma$. Modularity then follows from a short computation as the choice of coset representatives is permuted. The growth condition follows from that of f and the definition of F. This proves $F \in \mathcal{M}_{kn_\vartheta d_\Gamma}(\mathrm{SL}_2(\mathbb{Z}))$. $\square$

We can now state and prove the **valence formula**.

Theorem (Valence formula). *Suppose $f \in \mathcal{M}_k(\Gamma, \vartheta)$ is nonzero. Then*

$$\sum_{\mathfrak{a}\in\Gamma\backslash\widehat{\mathbb{Q}}} v_{\mathfrak{a}}(f) + \sum_{z\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}_\Gamma} \frac{2v_z(f)}{|\Gamma_z|} = \frac{kd_\Gamma}{12}.$$

Proof. We will first prove the valence formula for the modular group and then extend the result to Γ. Let $F \in \mathcal{M}_k(\mathrm{SL}_2(\mathbb{Z}))$ be nonzero so that its zeros are isolated. We claim that all of the zeros of F satisfy $\mathrm{Im}(z) < T$ for some positive T. To see this, let the Fourier series expansion of F be given by

$$F(z) = \sum_{n\geq 0} a(n)e(nz).$$

Note that F is bounded as $z \to \infty$ since it is holomorphic at ∞. Therefore, under the change of variables $e(z) \mapsto q$ this means

$$G(q) = \sum_{n\geq 0} a(n)q^n,$$

converges absolutely provided $|q| < 1$ and where $G(0) = \lim_{z\to\infty} F(z)$. Then G is holomorphic on the interior of the unit disk and the order of vanishing of G at 0 is exactly $v_\infty(F)$. Therefore we may write

$$G(q) = q^{v_\infty(F)}H(q),$$

where H is holomorphic on the interior of the unit disk and $H(0) \neq 0$. Whence there exists a positive r such that $G(q)$ is nonzero for $0 < |q| < r$. But then $F(z)$ is nonzero for $e^{-2\pi ny} < r$ which is to say $y > -\frac{\log(r)}{2\pi y}$. Taking $T > -\frac{\log(r)}{2\pi y}$ proves the claim.

It follows that every zero ρ of F has ordinate less than T. Moreover, $\gamma\rho$ is a zero for every $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ by modularity. For our choice of T, let $\eta = \sum_i \eta_i$ be the contour in Figure 2 where the horizontal line is along $t = T$ and the remaining contour is along the boundary of $\mathcal{F}$ except for indentations of radius ε around possible zeros of F along this boundary as well as the points i, ω, and $1+\omega$. Moreover, the paths are chosen to be symmetric with respect to the actions of S and T so that each zero ρ belonging to $\mathcal{F}$ has exactly one representative of its orbit occurring inside η except for when $\rho = i, \omega, 1+\omega$ which are excluded. As $v_z(F) = 0$ for every $z \in \mathcal{F} - \{i, \omega, 1+\omega\}$ that is not a zero of F, the argument principle gives

$$\frac{1}{2\pi i}\int_\eta \frac{F'}{F}(z)dz = \sideset{}{^*}\sum_{z\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}} \frac{2v_z(F)}{|\mathrm{SL}_2(\mathbb{Z})_z|},$$

where the $*$ indicates that we are excluding the orbits of i and ω. Now define integrals

$$I_i(\varepsilon) = \int_{\eta_i} \frac{F'}{F}(z)dz,$$

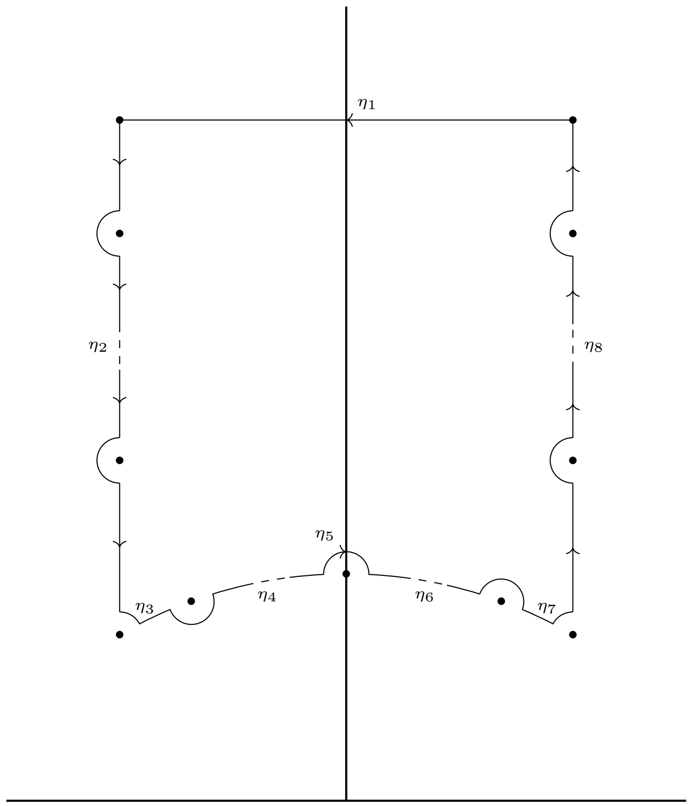

Figure 2: An indented standard fundamental domain contour.

so that

$$\frac{1}{2\pi i}\int_{\eta}\frac{F'}{F}(z)dz = \sum_i I_i(\varepsilon).$$

We will compute the integrals $I_i(\varepsilon)$ and then take the limit as $\varepsilon \to 0$. For the integrals along η_2 and η_8, modularity implies $F(z+1) = F(z)$ and observe that $T^{-1}\eta_8 = -\eta_2$. Whence

$$I_2(\varepsilon) + I_8(\varepsilon) = 0,$$

so that these integrals cancel. For the integral along η_1, the change of variables $e(z) \mapsto q$ takes η_1 to the circle $|q| = e^{-2\pi T}$ transversed once with negative orientation and $F(z)$ is taken to $G(q)$. Then from our previous discussion of $G(q)$, we have

$$I_1(\varepsilon) = -v_\infty(F).$$

The integrals along η_3 and η_7, as well as η_5, are computed in a similar manner. Let θ_ε be the angle such that η_3 and η_7 end and begin at angles $\frac{\pi}{6}+\theta_\varepsilon$ and $\frac{5\pi}{6}-\theta_\varepsilon$ relative to ω and $1+\omega$ respectively. For η_3, write $F(z) = (z-\omega)^{v_\omega(F)}F_\omega(z)$ where $F_\omega(z)$ holomorphic and nonzero at ω. Then

$$I_3(\varepsilon) = \frac{1}{2\pi i}\int_{\eta_4}\left(\frac{v_\omega(F)}{z-\omega} + \frac{F'_\omega}{F_\omega}(z)\right)dz,$$

and the change of variables $z \mapsto \omega + \varepsilon e^{i\theta}$ shows

$$I_3(\varepsilon) = -\frac{\frac{\pi}{3}-\theta_\varepsilon}{2\pi}v_\omega(F) + O(\varepsilon).$$

Since $\theta_\varepsilon \to 0$ as $\varepsilon \to 0$, it follows that

$$\lim_{\varepsilon \to 0} I_3(\varepsilon) = -\frac{v_\omega(F)}{|\mathrm{SL}_2(\mathbb{Z})_\omega|},$$

as $|\mathrm{SL}_2(\mathbb{Z})_\omega| = 6$ by Proposition 1.8. We argue similarly for η_7. Write $F(z) = (z-(1+\omega))^{v_{1+\omega}(F)} F_{1+\omega}(z)$ where $F_{1+\omega}(z)$ holomorphic and nonzero at $1+\omega$. Then

$$I_7(\varepsilon) = \frac{1}{2\pi i} \int_{\eta_7} \left(\frac{v_{1+\omega}(F)}{z-(1+\omega)} + \frac{F'_{1+\omega}}{F_{1+\omega}}(z) \right) dz,$$

and the change of variables $z \mapsto (1+\omega) + \varepsilon e^{i\theta}$ shows

$$I_7(\varepsilon) = \frac{\frac{\pi}{3} - \theta_\varepsilon}{2\pi} v_{1+\omega}(F) + O(\varepsilon).$$

Whence

$$\lim_{\varepsilon \to 0} I_7(\varepsilon) = -\frac{v_{1+\omega}(F)}{|\mathrm{SL}_2(\mathbb{Z})_{1+\omega}|},$$

where $|\mathrm{SL}_2(\mathbb{Z})_{1+\omega}| = 6$ by Proposition 1.8. For the integral along η_5, write $F(z) = (z-i)^{v_i(F)} F_i(z)$ where $F_i(z)$ is holomorphic and nonzero at i. Then

$$I_5(\varepsilon) = \frac{1}{2\pi i} \int_{\eta_5} \left(\frac{v_i(F)}{z-(i)} + \frac{F'_i}{F_i}(z) \right) dz,$$

and the change of variables $z \mapsto i + \varepsilon e^{i\theta}$ gives

$$I_5(\varepsilon) = -\frac{v_i(F)}{2} + O(\varepsilon).$$

Therefore

$$\lim_{\varepsilon \to 0} I_5(\varepsilon) = -\frac{2v_i(F)}{|\mathrm{SL}_2(\mathbb{Z})_i|},$$

as $|\mathrm{SL}_2(\mathbb{Z})_i| = 4$ by Proposition 1.8. For the integrals along η_4 and η_6, let C_ε be the finite union of segments along η_4 that lie on the boundary of $\mathcal{F}$. Modularity implies $F\left(-\frac{1}{z}\right) = z^k F(z)$ and notice that $S^{-1}\eta_6 = -\eta_4$. Hence

$$I_4(\varepsilon) + I_6(\varepsilon) = -\frac{1}{2\pi i} \int_{C_\varepsilon} \frac{k}{z}\, dz.$$

As $\varepsilon \to 0$, C_ε tends to the arc on the unit circle between radii $\frac{\pi}{2}$ and $\frac{2\pi}{3}$ with negative orientation. From the change of variables $z \mapsto e^{i\theta}$, we find that

$$\lim_{\varepsilon \to 0} (I_4(\varepsilon) + I_6(\varepsilon)) = \frac{k}{12}.$$

Altogether, this means

$$\lim_{\varepsilon \to 0} \frac{1}{2\pi i} \int_\eta \frac{F'}{F}(z) dz = \frac{k}{12} - v_\infty(F) - \frac{v_\omega(F)}{|\mathrm{SL}_2(\mathbb{Z})_\omega|} - \frac{v_{1+\omega}(F)}{|\mathrm{SL}_2(\mathbb{Z})_{1+\omega}|} - \frac{2v_i(F)}{|\mathrm{SL}_2(\mathbb{Z})_i|}, \tag{9}$$

whence

$$v_\infty(F) + \frac{v_\omega(F)}{|\mathrm{SL}_2(\mathbb{Z})_\omega|} + \frac{v_{1+\omega}(F)}{|\mathrm{SL}_2(\mathbb{Z})_{1+\omega}|} + \frac{2v_i(F)}{|\mathrm{SL}_2(\mathbb{Z})_i|} + \sideset{}{^*}\sum_{z\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}} \frac{2v_z(F)}{|\mathrm{SL}_2(\mathbb{Z})_z|} = \frac{k}{12}.$$

Since ω and $1+\omega$ are $\mathrm{SL}_2(\mathbb{Z})$-equivalent, $v_\omega(F) = v_{1+\omega}(F)$ and $|\mathrm{SL}_2(\mathbb{Z})_\omega| = |\mathrm{SL}_2(\mathbb{Z})_{1+\omega}|$ by Proposition 1.8. So the previous identity becomes

$$v_\infty(F) + \sum_{z\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}} \frac{2v_z(F)}{|\mathrm{SL}_2(\mathbb{Z})_z|} = \frac{k}{12},$$

which is precisely the claim for the modular group. It remains to prove the claim for Γ. So let $f \in \mathcal{M}_k(\Gamma,\vartheta)$ be nonzero. By Lemma 2.6 we may take

$$F(z) = \prod_{\beta\in\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})} ((f|_k\beta)(z))^{n_\vartheta},$$

and obtain

$$v_\infty(F) + \sum_{z\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}} \frac{2v_z(F)}{|\mathrm{SL}_2(\mathbb{Z})_z|} = \frac{kmd_\Gamma}{12}.$$

Observe that the Fourier series expansion of F is a product of those for f over a complete set of Γ-inequivalent cusps each with multiplicity m. Therefore the index of the first nonzero term in the Fourier series expansion of F is

$$v_\infty(F) = \sum_{\mathfrak{a}\in\Gamma\backslash\widehat{\mathbb{Q}}} mv_{\mathfrak{a}}(f).$$

Lastly, modularity implies $F(z) = 0$ if and only if $f(\beta z) = 0$ for some β in which case the zero is of multiplicity m. From this and the orbit-stabilizer theorem, we find that

$$\frac{2v_z(F)}{|\mathrm{SL}_2(\mathbb{Z})_z|} = \sum_{\beta\in\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})} \frac{2mv_{\beta z}(f)}{|\Gamma_{\beta z}|}.$$

But then Proposition 1.7 further implies

$$\sum_{z\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}} \frac{2v_z(F)}{|\mathrm{SL}_2(\mathbb{Z})_z|} = \sum_{z\in\Gamma\backslash\mathcal{F}_\Gamma} \frac{2mv_z(f)}{|\Gamma_z|}.$$

Substituting these identities into the formula for the modular group gives

$$\sum_{\mathfrak{a}\in\Gamma\backslash\widehat{\mathbb{Q}}} v_{\mathfrak{a}}(f) + \sum_{z\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}_\Gamma} \frac{2v_z(f)}{|\Gamma_z|} = \frac{kd_\Gamma}{12},$$

as desired. $\square$

With the valence formula in hand, it is not hard to show that $\mathcal{M}_k(\Gamma,\vartheta)$ is finite dimensional.

Theorem 2.7. *$\mathcal{M}_k(\Gamma,\vartheta)$ is finite dimensional. In particular, $\mathcal{S}_k(\Gamma,\vartheta)$ is too.*

Proof. Let $f \in \mathcal{M}_k(\Gamma,\vartheta)$. Recall that $v_{\mathfrak{a}}(f)$ and $v_z(f)$ are nonnegative. In particular, $v_\infty(f)$ is nonnegative and the valence formula implies $v_\infty(f) \le \frac{kd_\Gamma}{12}$ whenever f is nonzero. Let n be a positive integer such that $n \ge \frac{kd_\Gamma}{12}+1$ and consider the linear map $F : \mathcal{M}_k(\Gamma,\vartheta) \to \mathbb{C}^n$ sending f to its first n Fourier coefficients. Then F is injective since $v_\infty(f) < n$ and hence $\mathcal{M}_k(\Gamma,\vartheta)$ is finite dimensional. The second statement follows immediately. $\square$

2.4 The Petersson Inner Product

For further developments, we will require the Petersson inner product. Let Γ be a congruence subgroup and fix a nonnegative integer k. For any complex functions f and g on $\mathbb{H}$ such that $f(z)g(z)\mathrm{Im}(z)^k$ is Γ-invariant and bounded on $\mathbb{H}$, we define their **Petersson inner product** $\langle f,g\rangle$ by

$$\langle f,g\rangle = \frac{1}{V_\Gamma}\int_{\Gamma\backslash\mathbb{H}} f(z)\overline{g(z)}\mathrm{Im}(z)^k\,d\mu.$$

The integral is well-defined because the integrand is Γ-invariant and is absolutely uniformly convergent because it is bounded on $\mathbb{H}$ and Γ has finite covolume. An important property of the Petersson inner product is that it is invariant with respect to the slash operator provided the associated matrix normalizes Γ.

Proposition 2.8. *Let Γ be a congruence subgroup and suppose $\alpha \in \mathrm{GL}_2^+(\mathbb{R})$ normalizes Γ. Then*

$$\langle f|_k\alpha, g|_k\alpha\rangle = \langle f,g\rangle,$$

for any complex functions f and g on $\mathbb{H}$ such that $f(z)g(z)\mathrm{Im}(z)^k$ is Γ-invariant and bounded on $\mathbb{H}$.

Proof. This is just a computation. By definition, we have

$$\langle f|_k\alpha, g|_k\alpha\rangle = \frac{1}{V}\int_{\Gamma\backslash\mathbb{H}} (f|_k\alpha)(z)\overline{(g|_k\alpha)(z)}\mathrm{Im}(z)^k\,d\mu.$$

Making the change of variables $z \mapsto \alpha^{-1}z$, a short computation using normality, the cocycle condition, and Equation (5), shows that this integral becomes

$$\frac{1}{V}\int_{\Gamma\backslash\mathbb{H}} f(z)\overline{g(z)}\mathrm{Im}(z)^k\,d\mu = \langle f,g\rangle.$$

$\square$

We will primarily be interested in the case where $f,g \in \mathcal{M}_k(\Gamma\backslash\mathbb{H})$ and at least one of them is a cusp form. In this case, modularity implies $f(z)g(z)\mathrm{Im}(z)^k$ is Γ-invariant and is also bounded on $\mathbb{H}$ since at least one of f or g is a cusp form. The Petersson inner product makes $\mathcal{S}_k(\Gamma,\vartheta)$ into a complex Hilbert space. Indeed, it is clearly linear on $\mathcal{S}_k(\Gamma,\vartheta)$. Positive definiteness follows because

$$\langle f,f\rangle = \frac{1}{V_\Gamma}\int_{\Gamma\backslash\mathbb{H}} |f(z)|^2\mathrm{Im}(z)^k\,d\mu,$$

is nonnegative and zero if and only if f is identically zero. A short computation together using the fact that $\overline{d\mu} = d\mu$, which holds since $d\mu = \frac{dx\,dy}{y^2}$, proves conjugate symmetry. Whence $\mathcal{S}_k(\Gamma, \vartheta)$ is a complex inner product space with respect to the Petersson inner product. Since $\mathcal{S}_k(\Gamma, \vartheta)$ is finite dimensional by Theorem 2.7, it follows that $\mathcal{S}_k(\Gamma, \vartheta)$ is a complex Hilbert space.

2.5 Double Coset Operators

There are also linear operators that act on the space of modular forms. In fact, these operators allow us to transfer between modular forms on different congruence subgroups. They are defined via double cosets which we now study.

Let Γ_1 and Γ_2 be two congruence subgroups. Since congruence subgroups are preserved under intersection, it follows from Lemma 1.4 that $\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$ is also a congruence subgroup for any $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. In this setting, consider the double coset

$$\Gamma_1\alpha\Gamma_2 = \{\gamma_1\alpha\gamma_2 : \gamma_1 \in \Gamma_1 \text{ and } \gamma_2 \in \Gamma_2\}.$$

Note that Γ_1 acts on $\Gamma_1\alpha\Gamma_2$ by left multiplication. Whence $\Gamma_1\alpha\Gamma_2$ admits an orbit decomposition

$$\Gamma_1\alpha\Gamma_2 = \bigcup_{\beta \in \Gamma_1 \backslash \Gamma_1\alpha\Gamma_2} \Gamma_1\beta,$$

into right cosets. It turns out that this union is finite. To see this, we need to relate the orbit space $\Gamma_1\backslash\Gamma_1\alpha\Gamma_2$ to a quotient of Γ_2 with respect to the action of some subgroup. This will be the congruence subgroup $\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$. As this subgroup acts on Γ_2 by left multiplication, we again have an orbit space $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\backslash\Gamma_2$. These orbit spaces are related.

Lemma 2.9. *Let Γ_1 and Γ_2 be congruence subgroups and let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Then the right multiplication map*

$$\Gamma_2 \to \Gamma_1\backslash\Gamma_1\alpha\Gamma_2 \qquad \gamma_2 \mapsto \Gamma_1\alpha\gamma_2,$$

induces a bijection from $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\backslash\Gamma_2$ to $\Gamma_1\backslash\Gamma_1\alpha\Gamma_2$. In particular, if

$$\Gamma_2 = \bigcup_j (\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\gamma_2,$$

is a right coset decomposition then so is

$$\Gamma_1\alpha\Gamma_2 = \bigcup_j \Gamma_1\alpha\gamma_2.$$

Proof. The second statement is immediate from the first, so it suffices to show the first statement. The map is $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)$-invariant so that it induces a map on the quotient. Indeed, if $\gamma \in \alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$ then $\gamma = \alpha^{-1}\gamma_1\alpha$ for some $\gamma_1 \in \Gamma_1$. Thus $\Gamma_1\alpha\gamma\gamma_2 = \Gamma_1\alpha\gamma_2$ as claimed.

We now show that the induced map is a bijection. For injectivity, let γ_2 and γ_2' be representatives of orbits of $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\backslash\Gamma_2$ such that $\Gamma_1\alpha\gamma_2 = \Gamma_1\alpha\gamma_2'$. This implies

$\alpha\gamma_2(\gamma_2')^{-1} \in \Gamma_1\alpha$ and so $\gamma_2(\gamma_2')^{-1} \in \alpha^{-1}\Gamma_1\alpha$. But we also have $\gamma_2(\gamma_2')^{-1} \in \Gamma_2$ and these two facts together imply that $\gamma_2(\gamma_2')^{-1} \in \alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$. Hence

$$(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\gamma_2 = (\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\gamma_2',$$

from which injectivity follows. For surjectivity, any representative β of an orbit in $\Gamma_1\backslash\Gamma_1\alpha\Gamma_2$ is of the form $\beta = \gamma_1\alpha\gamma_2$ for some $\gamma_1 \in \Gamma_1$ and $\gamma_2 \in \Gamma_2$. But then γ_2 represents an orbit of $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\backslash\Gamma_2$ mapping to the orbit represented by β as $\Gamma_1\beta = \Gamma_1\alpha\gamma_2$. This shows surjectivity. □

With this lemma in hand, we can prove that the aforementioned right coset decomposition is finite.

Proposition 2.10. *Let Γ_1 and Γ_2 be congruence subgroups and let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Then the right coset decomposition*

$$\Gamma_1\alpha\Gamma_2 = \bigcup_{\beta\in\Gamma_1\backslash\Gamma_1\alpha\Gamma_2} \Gamma_1\beta,$$

is finite.

Proof. By Lemma 2.9, the number of orbits of $\Gamma_1\backslash\Gamma_1\alpha\Gamma_2$ is in bijection with those of $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\backslash\Gamma_2$ which is $[\Gamma_2 : \alpha^{-1}\Gamma_1\alpha \cap \Gamma_2]$. By Lemma 1.4, $\alpha^{-1}\Gamma_1\alpha \cap \mathrm{SL}_2(\mathbb{Z})$ is a congruence subgroup and thus $\alpha^{-1}\Gamma_1\alpha\cap\Gamma_2$ is as well. It follows that $[\Gamma_2 : \alpha^{-1}\Gamma_1\alpha\cap\Gamma_2] \le d_{\alpha^{-1}\Gamma_1\alpha\cap\Gamma_2}$ where $d_{\alpha^{-1}\Gamma_1\alpha\cap\Gamma_2}$ is finite which completes the proof. □

We are ready to introduce a class of general operators depending upon double cosets. First, we define an operator closely related to the slash operator. For any nonnegative integer k and any $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ we define the **double slash operator** $\|_k\alpha : C(\mathbb{H}) \to C(\mathbb{H})$ to be the linear operator given by

$$(f\|_k\alpha)(z) = \det(\alpha)^{k-1}j(\alpha, z)^{-k}f(\alpha z).$$

In the case of $\alpha \in \mathrm{SL}_2(\mathbb{R})$, the double slash operator takes the simplified form

$$(f\|_k\alpha)(z) = j(\alpha, z)^{-k}f(\alpha z).$$

The relationship to the slash operator is

$$(f\|_k\alpha)(z) = \det(\alpha)^{\frac{k}{2}-1}(f|_k\alpha)(z).$$

It follows that the double slash operator is multiplicative as we have

$$(f\|_k\alpha'\alpha)(z) = ((f\|_k\alpha')|_k\alpha)(z) \quad \text{and} \quad ((fg)\|_{k+\ell}\alpha)(z) = (f\|_k\alpha)(z)(g\|_\ell\alpha)(z),$$

for $f, g \in C(\mathbb{H})$, positive integers k and ℓ, and any $\alpha, \alpha' \in \mathrm{GL}_2^+(\mathbb{Q})$. Also, if $f \in \mathcal{M}_k(\Gamma, \vartheta)$ then modularity may be expressed as

$$(f\|_k\gamma)(z) = \vartheta(\gamma)f(z),$$

for all $\gamma \in \Gamma$. The reason for introducing the double slash operator is that it will be easier to work with when discussing Fourier coefficients of modular forms.

Now for any two congruence subgroups Γ_1 and Γ_2 and any $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$, we define the **double coset operator** $[\Gamma_1\alpha\Gamma_2]_k : \mathcal{M}_k(\Gamma_1) \to C(\mathbb{H})$ to be the linear operator given by

$$(f[\Gamma_1\alpha\Gamma_2]_k)(z) = \sum_{\beta \in \Gamma_1\backslash\Gamma_1\alpha\Gamma_2} (f\|_k\beta)(z).$$

This sum is finite by Proposition 2.10. To check that $f[\Gamma_1\alpha\Gamma_2]_k$ is well-defined amounts to showing that the sum is independent of the representatives β used. Indeed, if β and β' belong to the same orbit then $\beta' = \gamma\beta$ for some $\gamma \in \Gamma_1$. Modularity implies

$$(f\|_k\beta')(z) = (f\|_k\beta)(z),$$

which proves that $[\Gamma_1\alpha\Gamma_2]_k$ is well-defined. We will now show that the double coset operator takes $\mathcal{M}_k(\Gamma_1)$ into $\mathcal{M}_k(\Gamma_2)$ and preserves the subspaces of cusp forms.

Proposition 2.11. *For any two congruence subgroups Γ_1 and Γ_2, $[\Gamma_1\alpha\Gamma_2]_k$ maps $\mathcal{M}_k(\Gamma_1)$ into $\mathcal{M}_k(\Gamma_2)$ and preserves the subspaces of cusp forms.*

Proof. Let $f \in \mathcal{M}_k(\Gamma_1)$. Holomorphy is immediate since the sum in $f[\Gamma_1\alpha\Gamma_2]_k$ is finite. For modularity, let $\gamma \in \Gamma_2$. As β runs over a complete set of representatives for $\Gamma_1\backslash\Gamma_1\alpha\Gamma_2$ so does $\beta\gamma$. Modularity then follows from a short computation as the choice of coset representatives is permuted. For the growth condition, let $\sigma_\mathfrak{a}$ be a scaling matrix for a cusp $\mathfrak{a}$ of $\Gamma_2\backslash\mathbb{H}$. For any representative β, observe that $\beta\sigma_\mathfrak{a}\infty = \mathfrak{b}$ for some cusp $\mathfrak{b}$ of $\Gamma_1\backslash\mathbb{H}$. Then

$$((f[\Gamma_1\alpha\Gamma_2]_k)|_k\sigma_\mathfrak{a})(z) = \sum_{\beta \in \Gamma_1\backslash\Gamma_1\alpha\Gamma_2} \det(\beta)^{\frac{k}{2}-1}(f|_k\beta\sigma_\mathfrak{a})(z),$$

and so the growth condition follows from that of f. This proves the first statement. The second statement follows from this identity since $f[\Gamma_1\alpha\Gamma_2]_k$ is a cusp form if f is. $\square$

In the case $\Gamma_1 = \Gamma_2 = \Gamma$ for any congruence subgroup Γ, this result shows that the double coset operator $[\Gamma\alpha\Gamma]_k$ is a linear operator on $\mathcal{M}_k(\Gamma)$ and preserves the subspace $\mathcal{S}_k(\Gamma)$. Recalling that $\mathcal{S}_k(\Gamma)$ is a complex Hilbert space with respect to the Petersson inner product, this implies that $[\Gamma\alpha\Gamma]_k$ has an adjoint. We will show that the the adjoint is also a double coset operator whenever α normalizes Γ. To prove this we first require a lemma.

Lemma 2.12. *Suppose Γ is a congruence subgroup and let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Then there exists $\beta_j \in \mathrm{GL}_2^+(\mathbb{Q})$ such that $\Gamma\alpha\Gamma$ and $\Gamma\alpha^{-1}\Gamma$ admit right coset decompositions*

$$\Gamma\alpha\Gamma = \bigcup_j \Gamma\beta_j \quad \textit{and} \quad \Gamma\alpha^{-1}\Gamma = \bigcup_j \Gamma\beta_j^{-1}.$$

Proof. By Lemma 1.4 and that congruence subgroups are preserved under intersection, $\alpha\Gamma\alpha^{-1}\cap\Gamma$ and $\alpha^{-1}\Gamma\alpha\cap\Gamma$ are congruence subgroups. Clearly $[\Gamma : \alpha^{-1}\Gamma\alpha\cap\Gamma] = [\Gamma : \alpha\Gamma\alpha^{-1}\cap\Gamma]$

and so we can find representatives $\gamma_j \in \Gamma$ and $\gamma_j' \in \Gamma$ such that we have right coset decompositions

$$\Gamma = \bigcup_j (\alpha^{-1}\Gamma\alpha \cap \Gamma)\gamma_j \quad \text{and} \quad \Gamma = \bigcup_j (\alpha\Gamma\alpha^{-1} \cap \Gamma)(\gamma_j')^{-1}.$$

Invoking Lemma 2.9 twice, we obtain right coset decompositions

$$\Gamma\alpha\Gamma = \bigcup_j \Gamma\alpha\gamma_j \quad \text{and} \quad \Gamma\alpha^{-1}\Gamma = \bigcup_j \Gamma\alpha^{-1}(\gamma_j')^{-1}.$$

Also note that these right coset decompositions are inverses of each other. For each j, the right and left cosets $\Gamma\alpha\gamma_j$ and $\gamma_j'\alpha\Gamma$ have nonempty intersection. For if they did, for some j, we would have

$$\Gamma\alpha\gamma_j \subseteq \bigcup_{i \neq j} \gamma_i'\alpha\Gamma,$$

and thus

$$\Gamma\alpha\Gamma \subseteq \bigcup_{i \neq j} \gamma_i'\alpha\Gamma.$$

Upon inverting, this contradicts the previous right coset decomposition of $\Gamma\alpha^{-1}\Gamma$. Therefore we can find representatives $\beta_j \in \Gamma\alpha\gamma_j \cap \gamma_j'\alpha\Gamma$ for every j. Then we obtain right coset decompositions

$$\Gamma\alpha\Gamma = \bigcup_j \Gamma\beta_j \quad \text{and} \quad \Gamma\alpha^{-1}\Gamma = \bigcup_j \Gamma\beta_j^{-1}.$$

□

In order to describe the adjoint of $[\Gamma\alpha\Gamma]_k$ we will introduce some notation. Let

$$\alpha^* = \det(\alpha)\alpha^{-1},$$

be the adjugate of α for any $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. We will now describe the adjoint of $[\Gamma\alpha\Gamma]_k$ whenever α normalizes Γ. As remarked, this is another double coset operator.

Proposition 2.13. *Let Γ be a congruence subgroup and suppose $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ normalizes Γ. Then for any $f, g \in \mathcal{S}_k(\Gamma)$ we have*

$$\langle f|_k\alpha, g\rangle = \langle f, g|_k\alpha^{-1}\rangle \quad \textit{and} \quad \langle f\|_k\alpha, g\rangle = \langle f, g\|_k\alpha^*\rangle.$$

In particular,

$$[\Gamma\alpha\Gamma]_k^* = [\Gamma\alpha^*\Gamma]_k.$$

Proof. It follows from Proposition 2.8 that

$$\langle f|_k\alpha, g\rangle = \langle f, g|_k\alpha^{-1}\rangle \quad \text{and} \quad \langle f\|_k\alpha, g\rangle = \langle f, g\|_k\alpha^*\rangle,$$

where the second identity follows from the first by definition of the double slash operator. It remains to prove the adjoint formula for $[\Gamma\alpha\Gamma]_k^*$. By Lemma 2.12, we can find $\beta_j \in \mathrm{GL}_2^+(\mathbb{Q})$ such that

$$\Gamma\alpha\Gamma = \bigcup_j \Gamma\beta_j \quad \text{and} \quad \Gamma\alpha^{-1}\Gamma = \bigcup_j \Gamma\beta_j^{-1}.$$

Whence

$$\Gamma\alpha\Gamma = \bigcup_j \Gamma\beta_j \quad \text{and} \quad \Gamma\alpha^*\Gamma = \bigcup_j \Gamma\beta_j^*,$$

because $\det(\alpha) = \det(\beta_j)$. The adjoint formula follows at once from the first statement. □

3 Arithmetic of Modular Forms on Hecke Congruence Subgroups

If a is an integer modulo m with $(a, m) = 1$ we write a^{-1} for the inverse of a modulo m. For any complex z let

$$e(z) = e^{2\pi i z}.$$

Also, all sums and products over a quotient group will be consider to be over a complete set of representatives.

3.1 Poincaré Series and Eisenstein Series

We will now study the Poincaré series and Eisenstein Series on $\Gamma_0(N)\backslash\mathbb{H}$. As $\Gamma_0(N)$ is a Hecke congruence subgroup, $\Gamma_0(N)_\infty = B_{\Gamma_0(N)}$ and we may take $\sigma_\infty = I$. We will also take $\vartheta = \chi$ to be a Dirichlet character modulo N. Then for a cusp $\mathfrak{a}$ with scaling matrix $\sigma_{\mathfrak{a}}$, our series take the forms

$$P_{m,\chi,\mathfrak{a}}(z) = \sum_{\gamma\in\Gamma_0(N)_{\mathfrak{a}}\backslash\Gamma_0(N)} \overline{\chi}(\gamma)\left(e(m\,\cdot)\bigg|_k \sigma_{\mathfrak{a}}^{-1}\gamma\right)(z),$$

and

$$E_{\chi,\mathfrak{a}}(z) = \sum_{\gamma\in\Gamma_0(N)_{\mathfrak{a}}\backslash\Gamma_0(N)} \overline{\chi}(\gamma)\left(1\bigg|_k \sigma_{\mathfrak{a}}^{-1}\gamma\right)(z).$$

In either case, if χ is trivial or $\mathfrak{a} = \infty$ then we will suppress these dependencies accordingly. Also, recall that χ is always regular at ∞.

It turns out that we can explicitly compute the Fourier series expansions of the Poincaré series and Eisenstein series. Their Fourier coefficients involve Kloosterman sums, or more generally Salié sums when χ is not trivial, relative to cusps. This will essentially follow from a careful application of Poisson summation. The arguments involved are only slightly different. We first deduce the Fourier series expansion of the Poincaré series.

Proposition 3.1. *Let $\mathfrak{b}$ be a cusp of $\Gamma_0(N)\backslash\mathbb{H}$ with scaling matrix $\sigma_{\mathfrak{b}}$. The Fourier series expansion of $P_{m,\chi,\mathfrak{a}}(z)$ on $\Gamma_0(N)\backslash\mathbb{H}$ at the $\mathfrak{b}$ cusp is given by*

$$\begin{aligned}(P_{m,\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z) &= \delta_{\mathfrak{a},\mathfrak{b}}e(mz) \\ &\quad + \sum_{n\ge1}\left(2\pi i^{-k}\left(\frac{\sqrt{n}}{\sqrt{m}}\right)^{k-1}\sum_{c\in\mathcal{C}_{\mathfrak{a},\mathfrak{b}}}\frac{S_{\chi,\mathfrak{a},\mathfrak{b}}(m,n,c)}{c}J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c}\right)\right)e(nz),\end{aligned}$$

provided the index m is positive.

Proof. Write

$$(P_{m,\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z) = \sum_{\gamma\in\Gamma_0(N)_{\mathfrak{a}}\backslash\Gamma_0(N)} \overline{\chi}(\gamma)j(\sigma_{\mathfrak{a}}^{-1}\gamma\sigma_{\mathfrak{b}}, z)^{-k}e(m\sigma_{\mathfrak{a}}^{-1}\gamma\sigma_{\mathfrak{b}}z).$$

From the Bruhat decomposition for $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}$, any $\gamma \in \Gamma_0(N)_{\mathfrak{a}}\backslash\Gamma_0(N)$ is of the form $\gamma = \left(\begin{smallmatrix} a & b \\ c & c\ell+d \end{smallmatrix}\right)$ where $c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}$, $d \in \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)$, and $\ell \in \mathbb{Z}$. Furthermore, a short computation shows

$$\frac{az+b}{cz+(c\ell+d)} = \frac{a}{c} - \frac{1}{c^2z+c^2\ell+cd}.$$

Using these facts together with the periodicity of χ, we have

$$(P_{m,\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z) = \delta_{\mathfrak{a},\mathfrak{b}}e(mz) + \sum_{\substack{c\in\mathcal{C}_{\mathfrak{a},\mathfrak{b}} \\ d\in\mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c) \\ \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right)\in\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}}} \overline{\chi}(d)\sum_{\ell\in\mathbb{Z}} \frac{e\left(m\left(\frac{a}{c} - \frac{1}{c^2z+c^2\ell+cd}\right)\right)}{(cz+c\ell+d)^k}.$$

Now let

$$I_{c,d}(z) = \sum_{\ell\in\mathbb{Z}} \frac{e\left(m\left(\frac{a}{c} - \frac{1}{c^2z+c^2\ell+cd}\right)\right)}{(cz+c\ell+d)^k}.$$

We will apply the Poisson summation formula to $I_{c,d}(z)$. By the identity theorem it suffices to verify the claim for $z = iy$ with y positive. Accordingly, let $f(x)$ be given by

$$f(x) = \frac{e\left(m\left(\frac{a}{c} - \frac{1}{c^2(iy)+c^2x+cd}\right)\right)}{(c(iy)+cx+d)^k}.$$

Then $f(x)$ is Schwartz class. We will compute its Fourier transform

$$(\mathcal{F}f)(n) = \int_{-\infty}^{\infty} \frac{e\left(m\left(\frac{a}{c} - \frac{1}{c^2(iy)+c^2x+cd}\right)\right)}{(c(iy)+cx+d)^k}e(-nx)\,dx.$$

Complexify the integral to get

$$\int_{\mathrm{Im}(w)=0} \frac{e\left(m\left(\frac{a}{c} - \frac{1}{c^2(iy)+c^2w+cd}\right)\right)}{(c(iy)+cw+d)^k}e(-nw)\,dw,$$

and make the change of variables $w \mapsto w - \frac{d}{c} - (iy)$ to obtain

$$e\left(\frac{ma+nd}{c} + n(iy)\right)\int_{\mathrm{Im}(w)=y} \frac{e\left(-\frac{m}{c^2w}\right)}{(cw)^k}e(-nw)\,dw.$$

The integrand is meromorphic with a pole at $w = 0$ and has polynomial decay of order k provided n is nonpositive. So for these n, we conclude that the integral vanishes upon shifting the line of integration to $\mathrm{Im}(w) = T$ for positive T and taking the limit as $T \to \infty$. It remains to compute the integral for positive n. To do this, make the change of variables $w \mapsto -\frac{w}{2\pi in}$ to the last integral to rewrite it as

$$\frac{(-1)^k(2\pi in)^{k-1}}{c^k}\int_{(2\pi ty)} \frac{e^{w-\frac{4\pi^2mn}{c^2w}}}{w^k}\,dw.$$

Homotoping the line of integration to the Hankel contour about the negative real axis, we recognize the integral as the Schläflin integral representation for the J-Bessel function since m is positive. This gives

$$2\pi i^{-k}\left(\frac{\sqrt{n}}{\sqrt{m}}\right)^{k-1}\frac{1}{c}J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c}\right).$$

So in total we obtain

$$(\mathcal{F}f)(n)=\begin{cases}\left(2\pi i^{-k}\left(\frac{\sqrt{n}}{\sqrt{m}}\right)^{k-1}\frac{e\left(\frac{ma+nd}{c}\right)}{c}J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c}\right)\right)e\left(n(iy)\right) & \text{if } n\geq 1,\\ 0 & \text{if } n\leq 0.\end{cases}$$

Then by Poisson summation, we have

$$I_{c,r}(z)=\sum_{n\geq 1}\left(2\pi i^{-k}\left(\frac{\sqrt{n}}{\sqrt{m}}\right)^{k-1}\frac{e\left(\frac{ma+nd}{c}\right)}{c}J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c}\right)\right)e(nz).$$

Plugging this back into the Poincaré series, interchanging the sums over n and c by absolute convergence, and collecting the exponential sum over a and d, we at last obtain

$$\begin{aligned}(P_{m,\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z)&=\delta_{\mathfrak{a},\mathfrak{b}}e(mz)\\&\quad+\sum_{n\geq 1}\left(2\pi i^{-k}\left(\frac{\sqrt{n}}{\sqrt{m}}\right)^{k-1}\sum_{c\in\mathcal{C}_{\mathfrak{a},\mathfrak{b}}}\frac{S_{\chi,\mathfrak{a},\mathfrak{b}}(m,n,c)}{c}J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c}\right)\right)e(nz).\end{aligned}$$

which is our desired Fourier series expansion. □

An immediate consequence of this result is that the Poincaré series $P_{m,\chi,\mathfrak{a}}$ of positive index are cusp forms.

The argument to obtain the Fourier series expansion of the Eisenstein series is much the same. The only slight difference is that we will pass by the pole of the integrand instead of deforming to the Hankel contour about the negative real axis.

Proposition 3.2. *Let $\mathfrak{b}$ be a cusp of $\Gamma_0(N)\backslash\mathbb{H}$ with scaling matrix $\sigma_{\mathfrak{b}}$. The Fourier series expansion of $E_{\chi,\mathfrak{a}}(z)$ on $\Gamma_0(N)\backslash\mathbb{H}$ at the $\mathfrak{b}$ cusp is given by*

$$(E_{\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z)=\delta_{\mathfrak{a},\mathfrak{b}}+\sum_{n\geq 1}\left(\frac{(-2\pi i)^k n^{k-1}}{\Gamma(k)}\sum_{c\in\mathcal{C}_{\mathfrak{a},\mathfrak{b}}}\frac{S_{\chi}(0,n,c)}{c^k}\right)e(nz).$$

Proof. Write

$$(E_{\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z)=\sum_{\gamma\in\Gamma_0(N)_{\mathfrak{a}}\backslash\Gamma_0(N)}\overline{\chi}(\gamma)j(\sigma_{\mathfrak{a}}^{-1}\gamma\sigma_{\mathfrak{b}},z)^{-k}.$$

From the Bruhat decomposition for $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}$, any $\gamma\in\Gamma_0(N)_{\mathfrak{a}}\backslash\Gamma_0(N)$ is of the form $\gamma=\left(\begin{smallmatrix}a & b\\ c & c\ell+d\end{smallmatrix}\right)$ where $c\in\mathcal{C}_{\mathfrak{a},\mathfrak{b}}$, $d\in\mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)$, and $\ell\in\mathbb{Z}$. Furthermore, a short computation shows

$$\frac{az+b}{cz+(c\ell+d)}=\frac{a}{c}-\frac{1}{c^2z+c^2\ell+cd}.$$

Using these facts together with the periodicity of χ, we have

$$(E_{\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z) = \delta_{\mathfrak{a},\mathfrak{b}} + \sum_{\substack{c\in\mathcal{C}_{\mathfrak{a},\mathfrak{b}} \\ d\in\mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c) \\ \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right)\in\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}}} \overline{\chi}(d)\sum_{\ell\in\mathbb{Z}}\frac{1}{(cz+c\ell+d)^k}.$$

Now let

$$I_{c,d}(z) = \sum_{\ell\in\mathbb{Z}}\frac{1}{(cz+c\ell+d)^k}.$$

We will apply the Poisson summation formula to $I_{c,d}(z)$. By the identity theorem it suffices to verify the claim for $z = iy$ with y positive. Accordingly, let $f(x)$ be given by

$$f(x) = \frac{1}{(c(iy)+cx+d)^k}.$$

Then $f(x)$ is Schwartz class. We will compute its Fourier transform

$$(\mathcal{F}f)(n) = \int_{-\infty}^{\infty}\frac{1}{(c(iy)+cx+d)^k}e(-nx)\,dx.$$

Complexify the integral to get

$$\int_{\mathrm{Im}(w)=0}\frac{1}{(c(iy)+cw+d)^k}e(-nw)\,dw,$$

and make the change of variables $w \mapsto w - \frac{d}{c} - (iy)$ to obtain

$$e\left(\frac{nd}{c}+n(iy)\right)\int_{\mathrm{Im}(w)=y}\frac{1}{(cw)^k}e(-nw)\,dw.$$

The integrand is meromorphic with a pole at $w = 0$ and has polynomial decay of order k provided n is nonpositive. So for these n, we conclude that the integral vanishes upon shifting the line of integration to $\mathrm{Im}(w) = T$ for positive T and taking the limit as $T \to \infty$. It remains to compute the integral for positive n. To do this, make the change of variables $w \mapsto -\frac{w}{2\pi in}$ to the last integral to rewrite it as

$$\frac{(-1)^k(2\pi in)^{k-1}}{c^k}\int_{(2\pi ty)}\frac{e^w}{w^k}\,dw.$$

The integrand of the remaining integral is meromorphic with a pole of order k at $w = 0$. From the Laurent series expansion of $\frac{e^w}{w^k}$ about $w = 0$, the residue is easily seen to be $\Gamma(k)$. So upon shifting the line of integration to $(-\varepsilon)$, we pass by this pole and obtain

$$\frac{(-2\pi i)^kn^{k-1}}{\Gamma(k)c^k} + \frac{(-1)^k(2\pi in)^{k-1}}{c^k}\int_{(-\varepsilon)}\frac{e^w}{w^k}\,dw.$$

This last integrand has polynomial decay of order k. Whence we may again conclude that the integral vanishes upon shifting the line of integration to $\mathrm{Im}(w) = -T$ for positive T and taking the limit as $T \to \infty$. So in total we obtain

$$(\mathcal{F}f)(n) = \begin{cases} \left(\frac{(-2\pi i)^k t^{k-1}}{\Gamma(k)} \frac{e\left(\frac{nd}{c}\right)}{c^k}\right) e\left(n(iy)\right) & \text{if } n \geq 1, \\ 0 & \text{if } n \leq 0. \end{cases}$$

Then by Poisson summation, we have

$$I_{c,r}(z) = \sum_{n \geq 1} \left(\frac{(-2\pi i)^k n^{k-1}}{\Gamma(k)} \frac{e\left(\frac{nd}{c}\right)}{c^k}\right) e(nz).$$

Plugging this back into the Poincaré series, interchanging the sums over n and c by absolute convergence, and collecting the exponential sum over d, we at last obtain

$$(E_{\chi,\mathfrak{a}}|_k \sigma_{\mathfrak{b}})(z) = \delta_{\mathfrak{a},\mathfrak{b}} + \sum_{n \geq 1} \left(\frac{(-2\pi i)^k n^{k-1}}{\Gamma(k)} \sum_{c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}} \frac{S_\chi(0,n,c)}{c^k}\right) e(nz).$$

which is our desired Fourier series expansion. □

An interesting consequence is that the constant term in the Fourier series expansion of $E_{\chi,\mathfrak{a}}$ at the $\mathfrak{b}$ cusp is zero unless $\mathfrak{a}$ is Γ-equivalent to $\mathfrak{b}$. Therefore the Eisenstein series $E_{\chi,\mathfrak{a}}$ are actually never cusp forms. Accordingly, write $\mathcal{E}_k(\Gamma_0(N),\chi)$ for the subspace spanned by the Eisenstein series associated to a complete set of $\Gamma_0(N)$-inequivalent cusps. It turns out that $\mathcal{M}_k(\Gamma_0(N),\chi)$ decomposes as

$$\mathcal{M}_k(\Gamma_0(N),\chi) = \mathcal{S}_k(\Gamma_0(N),\chi) \oplus \mathcal{E}_k(\Gamma_0(N),\chi),$$

where $\mathcal{S}_k(\Gamma_0(N),\chi)$ is spanned by the Poincaré series of positive index. Moreover, this decomposition is orthogonal in the sense that the Petersson inner product between elements of $\mathcal{S}_k(\Gamma_0(N),\chi)$ and $\mathcal{E}_k(\Gamma_0(N),\chi)$ is zero. To show all of this, let $\mathfrak{a}$ be a cusp of $\Gamma_0(N)\backslash\mathbb{H}$ and choose a scaling matrix $\sigma_{\mathfrak{a}}$. Suppose $f \in \mathcal{S}_k(\Gamma_0(N),\chi)$ and let its Fourier series expansion at the $\mathfrak{a}$ cusp be

$$(f|_k \sigma_{\mathfrak{a}})(z) = \sum_{n \geq 1} a_{\mathfrak{a}}(n) e(nz).$$

For any positive integer m, define linear functionals $\phi_{m,\chi,\mathfrak{a}} : \mathcal{S}_k(\Gamma_0(N),\chi) \to \mathbb{C}$ by

$$\phi_{m,\chi,\mathfrak{a}}(f) = a_{\mathfrak{a}}(m).$$

Since $\mathcal{S}_k(\Gamma_0(N),\chi)$ is a finite dimensional complex Hilbert space, the Riesz representation theorem implies that there exists unique $v_{m,\chi,\mathfrak{a}} \in \mathcal{S}_k(\Gamma_0(N),\chi)$ such that

$$\langle f, v_{m,\chi,\mathfrak{a}} \rangle = a_{\mathfrak{a}}(m).$$

We would like to know what these cusp forms are. It turns out that $v_{m,\chi,\mathfrak{a}}(z)$ will be the Poincaré series $P_{m,\chi,\mathfrak{a}}(z)$ up to a constant.

Proposition 3.3. *Let $f \in \mathcal{S}_k(\Gamma_0(N), \chi)$ have Fourier coefficients $a_{\mathfrak{a}}(n)$ at the $\mathfrak{a}$ cusp of $\Gamma_0(N)\backslash\mathbb{H}$. Then*

$$\langle f, P_{m,\chi,\mathfrak{a}} \rangle = \frac{\Gamma(k-1)}{V_{\Gamma_0(N)}(4\pi m)^{k-1}} a_{\mathfrak{a}}(m),$$

for all positive integers m.

Proof. We will compute

$$\langle f, P_{m,\chi,\mathfrak{a}} \rangle = \frac{1}{V_{\Gamma_0(N)}} \int_{\Gamma_0(N)\backslash\mathbb{H}} f(z)\overline{P_{m,\chi,\mathfrak{a}}(z)} \mathrm{Im}(z)^k \, d\mu.$$

A short computation using Equation (6) along with modularity and the cocycle condition shows this integral may be expressed as

$$\frac{1}{V_{\Gamma_0(N)}} \int_{\Gamma_0(N)\backslash\mathbb{H}} \sum_{\gamma \in \Gamma_0(N)_{\mathfrak{a}}\backslash\Gamma_0(N)} \overline{j(\sigma_{\mathfrak{a}}^{-1}, \gamma z)^{-k}} f(\gamma z) e\left(-m\overline{\sigma_{\mathfrak{a}}^{-1}\gamma z}\right) \mathrm{Im}(\gamma z)^k \, d\mu.$$

Now unfold to obtain

$$\frac{1}{V_{\Gamma_0(N)}} \int_{\Gamma_0(N)_{\mathfrak{a}}\backslash\mathbb{H}} \overline{j(\sigma_{\mathfrak{a}}^{-1}, z)^{-k}} f(z) e\left(-m\overline{\sigma_{\mathfrak{a}}^{-1} z}\right) \mathrm{Im}(z)^k \, d\mu.$$

Making the change of variables $z \mapsto \sigma_{\mathfrak{a}} z$, another short computation using the fact that $\Gamma_0(N)_{\mathfrak{a}} = \sigma_{\mathfrak{a}}\Gamma_0(N)_\infty\sigma_{\mathfrak{a}}^{-1}$ together with Equation (6) and the cocycle condition shows that this integral becomes

$$\frac{1}{V_{\Gamma_0(N)}} \int_{\Gamma_0(N)_\infty\backslash\mathbb{H}} (f|_k\sigma_{\mathfrak{a}})(z) e(-m\overline{z}) \mathrm{Im}(z)^k \, d\mu.$$

Substituting in the Fourier series expansion of f at the $\mathfrak{a}$ cusp and interchanging the inner sum and integral by absolute convergence, a short computation gives

$$\frac{1}{V_{\Gamma_0(N)}} \int_0^\infty a_{\mathfrak{a}}(m) e^{-4\pi m y} y^k \, \frac{dy}{y^2},$$

in view of Equation (8), because every term integrates to zero except the m-th term. Performing the change of variables $y \mapsto \frac{y}{4\pi m}$, this latter integral is recognized as the gamma function and we obtain

$$\frac{\Gamma(k-1)}{V_{\Gamma_0(N)}(4\pi m)^{k-1}} a_{\mathfrak{a}}(m),$$

completing the proof. □

It follows immediately from this result that

$$v_{m,\chi,\mathfrak{a}}(z) = \frac{V_{\Gamma_0(N)}(4\pi m)^{k-1}}{\Gamma(k-1)} P_{m,\chi,\mathfrak{a}}(z).$$

Thus the Poincaré series $P_{m,\chi,\mathfrak{a}}(z)$ of positive index extract the Fourier coefficients of f at the $\mathfrak{a}$ cusp up to constants.

An analogous argument with Eisenstein series shows that they are orthogonal to cusp forms.

Proposition 3.4. *Let $f \in \mathcal{S}_k(\Gamma_0(N), \chi)$ have Fourier coefficients $a_{\mathfrak{a}}(n)$ at the $\mathfrak{a}$ cusp of $\Gamma_0(N)\backslash\mathbb{H}$. Then*

$$\langle f, E_{\chi,\mathfrak{a}}\rangle = 0.$$

Proof. We will compute

$$\langle f, E_{\chi,\mathfrak{a}}\rangle = \frac{1}{V_{\Gamma_0(N)}}\int_{\Gamma_0(N)\backslash\mathbb{H}} f(z)\overline{E_{\chi,\mathfrak{a}}(z)}\mathrm{Im}(z)^k\,d\mu.$$

A short computation using Equation (6) along with modularity and the cocycle condition shows this integral may be expressed as

$$\frac{1}{V_{\Gamma_0(N)}}\int_{\Gamma_0(N)\backslash\mathbb{H}}\sum_{\gamma\in\Gamma_0(N)_{\mathfrak{a}}\backslash\Gamma_0(N)}\overline{j(\sigma_{\mathfrak{a}}^{-1},\gamma z)^{-k}}f(\gamma z)\mathrm{Im}(\gamma z)^k\,d\mu.$$

Now unfold to obtain

$$\frac{1}{V_{\Gamma_0(N)}}\int_{\Gamma_0(N)_{\mathfrak{a}}\backslash\mathbb{H}}\overline{j(\sigma_{\mathfrak{a}}^{-1},z)^{-k}}f(z)\mathrm{Im}(z)^k\,d\mu.$$

Making the change of variables $z\mapsto\sigma_{\mathfrak{a}}z$, another short computation using the fact that $\Gamma_0(N)_{\mathfrak{a}} = \sigma_{\mathfrak{a}}\Gamma_0(N)_\infty\sigma_{\mathfrak{a}}^{-1}$ together with Equation (6) and the cocycle condition shows that this integral becomes

$$\frac{1}{V_{\Gamma_0(N)}}\int_{\Gamma_0(N)_\infty\backslash\mathbb{H}}(f|_k\sigma_{\mathfrak{a}})(z)\mathrm{Im}(z)^k\,d\mu.$$

Substituting in the Fourier series expansion of f at the $\mathfrak{a}$ cusp and interchanging the inner sum and integral by absolute convergence, a short computation gives

$$\frac{1}{V_{\Gamma_0(N)}}\int_0^\infty a_{\mathfrak{a}}(0)y^k\,\frac{dy}{y^2},$$

in view of Equation (8), because every term integrates to zero except the constant term. This integral is zero because f is a cusp form which completes the proof. ☐

With these results in hand we can prove that the Poincaré series of positive index span the space of cusp forms and our desired decomposition.

Theorem 3.5. *The Poincaré series $P_{m,\chi}(z)$ of positive index span $\mathcal{S}_k(\Gamma_0(N),\chi)$. Moreover, we have the direct sum decomposition*

$$\mathcal{M}_k(\Gamma_0(N),\chi) = \mathcal{S}_k(\Gamma_0(N),\chi)\oplus\mathcal{E}_k(\Gamma_0(N),\chi),$$

which is orthogonal in the sense that

$$\langle P_{m,\chi,\mathfrak{b}}, E_{\chi,\mathfrak{a}}\rangle = 0.$$

Proof. Let $f \in \mathcal{S}_k(\Gamma_0(N), \chi)$. Proposition 3.3 shows that $\langle f, P_{m,\chi,\mathfrak{a}} \rangle = 0$ if and only if the m-th Fourier coefficient of f is zero. So f is orthogonal to all of the Poincaré series $P_{m,\chi}$ of positive index if and only if every Fourier coefficient is zero. But this happens if and only if f is identically zero. This proves the first statement.

To prove the direct sum decomposition, we first show that $\mathcal{E}_k(\Gamma_0(N), \chi)$ is disjoint from $\mathcal{S}_k(\Gamma_0(N), \chi)$. Indeed, suppose there are constants $a_{\mathfrak{a}}$ such that

$$\sum_{\mathfrak{a} \in \Gamma_0(N) \backslash \widehat{\mathbb{Q}}} a_{\mathfrak{a}} E_{\chi,\mathfrak{a}}(z),$$

is a cusp form. Recall from Proposition 3.2 that the constant term in the Fourier series expansion of $E_{\chi,\mathfrak{a}}$ at the $\mathfrak{b}$ cusp is $\delta_{\mathfrak{a},\mathfrak{b}}$. Therefore $a_{\mathfrak{a}} = 0$ which proves that these spaces are disjoint. Moreover, this fact also implies that

$$f(z) - \sum_{\mathfrak{a} \in \Gamma_0(N) \backslash \widehat{\mathbb{Q}}} a_{\mathfrak{a}}(n) E_{\chi,\mathfrak{a}}(z),$$

is a cusp form for any $f \in \mathcal{M}_k(\Gamma_0(N), \chi)$ with Fourier coefficients $a_{\mathfrak{a}}(n)$ at the $\mathfrak{a}$ cusp. The direct sum decomposition follows at once. Lastly, the orthogonality statement is immediate from Proposition 3.4 since the Poincaré series are cusp forms. □

3.2 Diamond and Hecke Operators

The diamond and Hecke operators are sums of double coset operators. They will be used to construct a linear theory of modular forms on $\Gamma_0(N)\backslash\mathbb{H}$ via their Fourier coefficients. Actually, these operators will be defined for modular forms on $\Gamma_1(N)\backslash\mathbb{H}$ with trivial nebentypus. However, it turns out that such modular forms arise from those on $\Gamma_0(N)\backslash\mathbb{H}$ with character via the diamond operators. So we will discuss the diamond operators first and the Hecke operators second.

To define the diamond operators, we need to consider the congruence subgroups $\Gamma_1(N)$ and $\Gamma_0(N)$. Recall that $\Gamma_1(N) \subseteq \Gamma_0(N)$ and consider the surjective homomorphism

$$\Gamma_0(N) \to (\mathbb{Z}/N\mathbb{Z})^* \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto d \pmod{N}.$$

Its kernel is exactly $\Gamma_1(N)$ so that $\Gamma_1(N)$ is a normal subgroup of $\Gamma_0(N)$ whence

$$\Gamma_0(N)/\Gamma_1(N) \cong (\mathbb{Z}/N\mathbb{Z})^*.$$

In fact, $\Gamma_0(N)$ admits the right coset decomposition

$$\Gamma_0(N) = \bigcup_{\substack{d \pmod{N} \\ (d,N)=1}} \Gamma_1(N)\gamma_d, \tag{10}$$

where $\gamma_d \in \Gamma_0(N)$ is any matrix satisfying $\gamma_d \equiv \left(\begin{smallmatrix} d^{-1} & 0 \\ 0 & d \end{smallmatrix}\right) \pmod{N}$. By surjectivity, a right coset decomposition exists so it only suffices to show that the representatives γ_d have this particular form. Any $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$ satisfies $\gamma \equiv \left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) \pmod{N}$ so that d is the inverse of a modulo N. As $(d, N) = 1$, there exists an integer n such that $nd \equiv -b \pmod{N}$. Then

$$\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \gamma \equiv \begin{pmatrix} d^{-1} & 0 \\ 0 & d \end{pmatrix} \pmod{N},$$

and so we can use the representatives γ_d. Since $\Gamma_1(N)$ is norm in $\Gamma_0(N)$, this right coset decomposition is also a left coset decomposition. Moreover, every $\gamma \in \Gamma_1(N)\gamma_d$ is of the form $\gamma \equiv \left(\begin{smallmatrix} * & * \\ 0 & d \end{smallmatrix}\right) \pmod{N}$ in view of

$$\begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \gamma_d \equiv \begin{pmatrix} * & * \\ 0 & d \end{pmatrix} \pmod{N},$$

which is to say that they have the same bottom right entry modulo N.

Let d be a positive integer. We will use the representatives γ_d to define the diamond operators. The **diamond operator** $\langle d \rangle : \mathcal{M}_k(\Gamma_1(N)) \to \mathcal{M}_k(\Gamma_1(N))$ is defined to be the double coset operator given by

$$(\langle d \rangle f)(z) = \begin{cases} (f[\Gamma_1(N)\gamma_d\Gamma_1(N)]_k)(z) & \text{if } (d, N) = 1, \\ 0 & \text{if } (d, N) > 1. \end{cases}$$

The diamond operators are well-defined in view of Equation (10). Moreover, there is only one orbit representative in double coset when $(d, N) = 1$. Indeed, normality of $\Gamma_1(N)$ in $\Gamma_0(N)$ implies $\Gamma_1(N)\gamma_d\Gamma_1(N) = \Gamma_1(N)\gamma_d$.

$$(\langle d \rangle f)(z) = (f\|_k\gamma_d)(z).$$

In any case, we find from the observation

$$\begin{pmatrix} a^{-1} & 0 \\ 0 & a \end{pmatrix} \begin{pmatrix} d^{-1} & 0 \\ 0 & d \end{pmatrix} \equiv \begin{pmatrix} (ad)^{-1} & 0 \\ 0 & ad \end{pmatrix} \pmod{N},$$

that the diamond operators are multiplicative, commutative, and invertible with $\langle d^* \rangle$ being the inverse of $\langle d \rangle$ where d^* is any positive integer satisfying $d^*d \equiv 1 \pmod{N}$ whenever $(d, N) = 1$. In particular, the diamond operators induce an action of $\Gamma_0(N)$ on $\mathcal{M}_k(\Gamma_1(N))$, and since $\Gamma_1(N)$ acts trivially and is normal in $\Gamma_0(N)$, there is a further induced action of $(\mathbb{Z}/N\mathbb{Z})^*$ in view of the previous isomorphism.

One reason the diamond operators are useful is that they decompose $\mathcal{M}_k(\Gamma_1(N))$ into eigenspaces. For any Dirichlet character χ modulo N, we let

$$\mathcal{M}_k(N, \chi) = \{f \in \mathcal{M}_k(\Gamma_1(N)) : \langle d \rangle f = \chi(d)f\},$$

be the χ-eigenspace. Also let $\mathcal{S}_k(N, \chi)$ be the corresponding subspace of cusp forms. In the case χ is trivial we will suppress this dependence. Then $\mathcal{M}_k(\Gamma_1(N))$ admits a decomposition into these eigenspaces.

Proposition 3.6. $\mathcal{M}_k(\Gamma_1(N))$ *admits the direct sum decomposition*

$$\mathcal{M}_k(\Gamma_1(N)) = \bigoplus_{\chi \pmod{N}} \mathcal{M}_k(N, \chi).$$

Moreover, this decomposition respects the subspace of cusp forms.

Proof. The diamond operators induce a representation Φ of $(\mathbb{Z}/N\mathbb{Z})^*$ over $\mathcal{M}_k(\Gamma_1(N))$. Explicitly, this representation is

$$\Phi : (\mathbb{Z}/N\mathbb{Z})^* \times \mathcal{M}_k(\Gamma_1(N)) \to \mathcal{M}_k(\Gamma_1(N)) \qquad (d, f) \mapsto \langle d \rangle f.$$

Since $(\mathbb{Z}/N\mathbb{Z})^*$ is a finite abelian group, every complex representation is completely reducible and every irreducible representation is one-dimensional. As the characters of $(\mathbb{Z}/N\mathbb{Z})^*$ are the Dirichlet characters modulo N, the decomposition as a direct sum follows. Moreover, it preserves the subspace of cusp forms since Φ restricts to a representation of $(\mathbb{Z}/N\mathbb{Z})^*$ over $\mathcal{S}_k(\Gamma_1(N))$. $\square$

Observe that every matrix in $\gamma \in \Gamma_0(N)$ is of the form $\gamma \equiv \left(\begin{smallmatrix} * & * \\ 0 & d \end{smallmatrix}\right) \pmod N$ for some integer d with $(d, N) = 1$. So this result shows that the diamond operators sieve modular forms on $\Gamma_1(N)\backslash\mathbb{H}$ with trivial character in terms of modular forms on $\Gamma_0(N)\backslash\mathbb{H}$ with nontrivial characters. In particular, we have

$$\mathcal{M}_k(N, \chi) = \mathcal{M}_k(\Gamma_0(N), \chi) \quad \text{and} \quad \mathcal{S}_k(N, \chi) = \mathcal{S}_k(\Gamma_0(N), \chi).$$

We are now ready to discuss the Hecke operators. First, for any positive integer m, set

$$\Delta_m = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{Mat}_2^m(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & * \end{pmatrix} \pmod N \right\}.$$

In other words, Δ_m is subset of $\mathrm{GL}_2^+(\mathbb{Q})$ consisting of determinant m matrices α with integral entries such that $\alpha \equiv \left(\begin{smallmatrix} 1 & * \\ 0 & * \end{smallmatrix}\right) \pmod N$. From the equivalence

$$\begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & * \\ 0 & * \end{pmatrix} \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & * \end{pmatrix} \pmod N,$$

it follows that $\Gamma_1(N)$ acts on Δ_m by both left and right multiplication. Whence Δ_m admits an orbit decomposition

$$\Delta_m = \bigcup_{\alpha \in \Gamma_1(N)\backslash\Delta_m/\Gamma_1(N)} \Gamma_1(N)\alpha\Gamma_1(N),$$

into double cosets. To see that this decomposition is finite, write α in smith normal form as $\alpha = \gamma \left(\begin{smallmatrix} a & 0 \\ 0 & d \end{smallmatrix}\right) \gamma'$ for some $\gamma, \gamma' \in \mathrm{SL}_2(\mathbb{Z})$ and positive integers a and d with $ad = m$ and $a \mid d$. As $\Gamma_1(N)$ has finite index in $\mathrm{SL}_2(\mathbb{Z})$, and there are finitely many choices for a and d, we conclude that this double coset decomposition is finite. We define the m-th **Hecke operator** $T_m : \mathcal{M}_k(\Gamma_1(N)) \to \mathcal{M}_k(\Gamma_1(N))$ to be the linear operator given by

$$(T_m f)(z) = \sum_{\alpha \in \Gamma_1(N)\backslash\Delta_m/\Gamma_1(N)} [\Gamma_1(N)\alpha\Gamma_1(N)]_k f(z).$$

The Hecke operator T_m is well-defined because it only depends the double coset decomposition of Δ_m. In fact, we can find an explicit set of double coset representatives for Δ_m.

Proposition 3.7. *There is a double coset decomposition*

$$\Delta_m = \bigcup_{\substack{ad=m\\ a|d\\ (a,N)=1}} \bigcup_{b \pmod d} \Gamma_1(N)\gamma_a \begin{pmatrix} a & 0\\ 0 & d\end{pmatrix} \Gamma_1(N).$$

Proof. The reverse inclusion holds because $\Gamma_1(N)$ acts on Δ_m by left and right multiplication and the representatives $\gamma_a \left(\begin{smallmatrix} a & 0\\ 0 & d\end{smallmatrix}\right)$ are elements of Δ_m in view of

$$\gamma_a \begin{pmatrix} a & 0\\ 0 & d\end{pmatrix} \equiv \begin{pmatrix} 1 & *\\ 0 & *\end{pmatrix} \pmod N.$$

For the forward inclusion, let L be the set of 2×1 column vectors with entries in $\mathbb{Z}$ which is a finite generated abelian group. Set

$$L_0 = \left\{ \begin{pmatrix} a\\ b\end{pmatrix} \in L : b \equiv 0 \pmod N \right\}.$$

Observe that L is a $\mathrm{Mat}_2(\mathbb{Z})$-module by left matrix multiplication, and while L_0 is not a $\mathrm{Mat}_2(\mathbb{Z})$-submodule, α preserves L_0 in view of

$$\begin{pmatrix} 1 & *\\ 0 & *\end{pmatrix} \begin{pmatrix} *\\ 0\end{pmatrix} \equiv \begin{pmatrix} *\\ 0\end{pmatrix} \pmod N.$$

Since αL has index m in L, and L_0 has index N in L, together we find

$$[L : \alpha L_0] = Nm.$$

As αL_0 is a finitely generated abelian group, the structure theorem for finitely generated abelian groups implies the existence of a basis u, v of L and positive integers a and b satisfying $ab = Nm$ and $a \mid b$ such that

$$\alpha L_0 = au\mathbb{Z} + bv\mathbb{Z}.$$

We will now show $(a, N) = 1$. If not then $\alpha L_0 \subseteq pL$ and since $\left(\begin{smallmatrix} 1\\ 0\end{smallmatrix}\right) \in L_0$ this would imply that the first row of α is divisible by p which is impossible as $\alpha \equiv \left(\begin{smallmatrix} 1 & *\\ 0 & *\end{smallmatrix}\right) \pmod N$. Whence $(a, N) = 1$ so that $a \mid m$ and $b \mid N$. Then $u\mathbb{Z} + Nv\mathbb{Z}$ is the unique index N subgroup of L containing αL_0 and so

$$L_0 = u\mathbb{Z} + Nv\mathbb{Z}.$$

Setting $d = \frac{b}{N}$, we see that $au\mathbb{Z} + dv\mathbb{Z}$ is the unique index m subgroup of L containing αL_0 so that

$$\alpha L = au\mathbb{Z} + dv\mathbb{Z}.$$

Now write $u = \left(\begin{smallmatrix} u_1\\ u_2\end{smallmatrix}\right)$ and $v = \left(\begin{smallmatrix} v_1\\ v_2\end{smallmatrix}\right)$. Define $\gamma, \gamma' \in \mathrm{Mat}_2(\mathbb{Z})$ by

$$\gamma = \begin{pmatrix} u_1 & v_1\\ u_2 & v_2\end{pmatrix} \quad \text{and} \quad \gamma' = \begin{pmatrix} a & 0\\ 0 & d\end{pmatrix}^{-1} \gamma^{-1}\alpha,$$

so that

$$\alpha = \gamma \begin{pmatrix} a & 0\\ 0 & d\end{pmatrix} \gamma'.$$

As u, v are a basis for L, replacing v by $-v$ if necessary, we may ensure $\gamma \in \mathrm{SL}_2(\mathbb{Z})$. Then a short computation shows $\gamma' \in \mathrm{SL}_2(\mathbb{Z})$. Also, as $u \in L_0$ and $au \in \alpha L_0$, we actually have $\gamma, \gamma' \in \Gamma_0(N)$. It follows from Equation (10) and normality of $\Gamma_0(N)$ that we may write

$$\alpha = \eta \gamma_s \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \gamma_t \eta',$$

for some $\eta, \eta' \in \Gamma_1(N)$ and s and t modulo N with $(st, N) = 1$. Furthermore, as $\Gamma_1(N)\gamma_s\gamma_t = \Gamma_1(N)\gamma_{st}$ and

$$\gamma_t^{-1} \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \gamma_t \equiv \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \pmod{N},$$

we may further write

$$\alpha = \eta \gamma_{st} \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \eta',$$

for some potentially different choices of η and η'. As $\alpha \in \Delta_m$, this identity implies

$$\begin{pmatrix} 1 & * \\ 0 & * \end{pmatrix} \equiv \begin{pmatrix} a(st)^{-1} & * \\ 0 & dst \end{pmatrix} \pmod{N}.$$

Whence $st \equiv a \pmod{N}$. Therefore

$$\alpha = \eta \gamma_a \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \eta',$$

proving the forward inclusion. It remains to show that the union is disjoint. If not, there exist $\eta_1, \eta_2 \in \Gamma_1(N)$ such that

$$\eta_1 \gamma_{a_1} \begin{pmatrix} a_1 & 0 \\ 0 & d_1 \end{pmatrix} = \gamma_{a_2} \begin{pmatrix} a_2 & 0 \\ 0 & d_2 \end{pmatrix} \eta_2,$$

for matrices $\gamma_{a_i} \left(\begin{smallmatrix} a_i & b_i \\ 0 & d_i \end{smallmatrix}\right)$ representing distinct double cosets and satisfying $a_i d_i = m$, $a_i \mid d_i$, and $(a_i, N) = 1$. But as the left-hand and right-hand sides are smith normal form, we conclude that $a_1 = a_2$ and $d_1 = d_2$. But then $\gamma_{a_i} \left(\begin{smallmatrix} a_i & b_i \\ 0 & d_i \end{smallmatrix}\right)$ represent the same double coset which is a contradiction proving the union is disjoint. $\square$

From this double coset decomposition, the Hecke operators can be expressed as

$$(T_m f)(z) = \sum_{\substack{ad=m \\ a \mid d \\ (a,N)=1}} \left[\Gamma_1(N) \gamma_a \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \Gamma_1(N) \right]_k f(z). \tag{11}$$

In order to compute the Hecke operators explicitly we need to find a complete set of representatives for each of the double cosets $\Gamma_1(N)\backslash\Gamma_1(N)\alpha\Gamma_1(N)$ in the sum. But this is the same as finding a complete set of representatives for $\Gamma_1(N)\backslash\Delta_m$ which is much easier to accomplish.

Proposition 3.8. *There is an right coset decomposition*

$$\Delta_m = \bigcup_{\substack{ad=m \\ (a,N)=1}} \bigcup_{b \pmod{d}} \Gamma_1(N)\gamma_a \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}.$$

Proof. The reverse inclusion holds because $\Gamma_1(N)$ acts on Δ_m by left and right multiplication and the representatives $\gamma_a \left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right)$ are elements of Δ_m in view of

$$\gamma_a \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & * \end{pmatrix} \pmod{N}.$$

For the forward inclusion, let $\alpha \in \Delta_m$. Putting α into Hermite normal form, we may write

$$\alpha = \gamma \begin{pmatrix} a & b \\ 0 & d \end{pmatrix},$$

for some $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and positive integers a, b, and d with $ad = m$ and b modulo d. As $\alpha \in \Delta_m$, this identity implies

$$\begin{pmatrix} 1 & * \\ 0 & m \end{pmatrix} \equiv \gamma \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \pmod{N}.$$

Then $(a, N) = 1$ as γ necessarily satisfies $\gamma \equiv \left(\begin{smallmatrix} a^{-1} & * \\ 0 & a \end{smallmatrix}\right) \pmod{N}$ and so $\gamma \in \Gamma_0(N)$. By Equation (10), we may write

$$\alpha = \eta\gamma_a \begin{pmatrix} a & b \\ 0 & d \end{pmatrix},$$

proving the forward inclusion. It remains to show that the union is disjoint. If not, then there exist $\eta_1, \eta_2 \in \Gamma_1(N)$ satisfying

$$\eta_1\gamma_{a_1} \begin{pmatrix} a_1 & b_1 \\ 0 & d_1 \end{pmatrix} = \eta_2\gamma_{a_2} \begin{pmatrix} a_2 & b_2 \\ 0 & d_2 \end{pmatrix},$$

for matrices $\gamma_{a_i} \left(\begin{smallmatrix} a_i & b_i \\ 0 & d_i \end{smallmatrix}\right)$ representing distinct right cosets and satisfying $a_i d_i = m$, $(a_i, N) = 1$, and b_i modulo d_i. Then

$$\begin{pmatrix} a_1 & b_1 \\ 0 & d_1 \end{pmatrix} \begin{pmatrix} a_2 & b_2 \\ 0 & d_2 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{a_1}{a_2} & * \\ 0 & \frac{d_1}{d_2} \end{pmatrix},$$

belongs to $\mathrm{SL}_2(\mathbb{Z})$ which forces $a_1 = a_2$ and $d_1 = d_2$. Whence this computation becomes

$$\begin{pmatrix} a_1 & b_1 \\ 0 & d_1 \end{pmatrix} = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a_2 & b_2 \\ 0 & d_2 \end{pmatrix},$$

for some integer t. As b_1 and b_2 are both modulo $d_1 = d_2$, we must have $t = 0$ and so $b_1 = b_2$. But then the matrices $\gamma_{a_i} \left(\begin{smallmatrix} a_i & b_i \\ 0 & d_i \end{smallmatrix}\right)$ represent the same right coset which is a contradiction. Therefore the union is disjoint. $\square$

From this right coset decomposition, we immediately find that the Hecke operators may be expressed in the form

$$(T_m f)(z) = \sum_{ad=m} \sum_{b \pmod{d}} \left((\langle a \rangle f) \bigg\|_k \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \right)(z), \tag{12}$$

where we are permitted to remove the condition $(a, N) = 1$ since the diamond operator $\langle a \rangle$ acts as the zero operator whenever $(a, N) > 1$. This formula can further be expressed explicitly as

$$(T_m f)(z) = m^{k-1} \sum_{ad=m} \frac{1}{d^k} \sum_{b \pmod{d}} (\langle a \rangle f) \left(\frac{az+b}{d} \right). \tag{13}$$

Since the diamond operators are commutative, this latter formula shows that they commute with the Hecke operators. In view of Proposition 3.6, it follows that the Hecke operators preserve the χ-eigenspaces. This is to say that the Hecke operators restrict to linear operators on $\mathcal{M}_k(N,\chi)$ and $\mathcal{S}_k(N,\chi)$ respectively. We can also use this formula to deduce how the Hecke operators act on the Fourier series expansions of modular forms in these subspaces.

Proposition 3.9. *Let $f \in \mathcal{M}_k(\Gamma_1(N))$ have Fourier series expansion*

$$f(z) = \sum_{n \geq 0} a(n) e(nz),$$

Then, for any positive integer m, the Fourier series expansion of $T_m f$ is

$$(T_m f)(z) = \sum_{n \geq 0} \left(\sum_{d \mid (n,m)} \langle d \rangle d^{k-1} a\left(\frac{nm}{d^2} \right) \right) e(nz).$$

In particular, if $f \in \mathcal{M}_k(N,\chi)$ then the Fourier series expansion of $T_m f$ is

$$(T_m f)(z) = \sum_{n \geq 0} \left(\sum_{d \mid (n,m)} \chi(d) d^{k-1} a\left(\frac{nm}{d^2} \right) \right) e(nz).$$

Proof. In view of Proposition 3.6, it suffices assume $f \in \mathcal{M}_k(N,\chi)$ and prove the second formula. By Equation (13), write

$$(T_m f)(z) = m^{k-1} \sum_{ad=m} \frac{\chi(a)}{d^k} \sum_{b \pmod{d}} \sum_{n \geq 0} a(n) e\left(\frac{anz + bn}{d} \right).$$

For fixed d, the sum over b is

$$\sum_{b \pmod{d}} e\left(\frac{bn}{d} \right) = \begin{cases} d & \text{if } d \mid n, \\ 0 & \text{if } d \nmid n, \end{cases}$$

because in the first case every term is 1 and in the second it is a geometric sum with common ratio $e\left(\frac{n}{d}\right)$. Whence our expression becomes

$$\sum_{ad=m} a^{k-1}\chi(a)\sum_{\substack{n\geq 0\\ d\mid n}} a(n)e\left(\frac{anz}{d}\right),$$

since $ad=m$. The change of variables $n\mapsto \frac{nd}{a}$ yields

$$\sum_{ad=m}\sum_{\substack{n\geq 0\\ a\mid n}} a^{k-1}\chi(a)a\left(\frac{nd}{a}\right)e(nz).$$

Now write $d=\frac{m}{a}$ to obtain

$$\sum_{n\geq 0}\left(\sum_{a\mid(n,m)}\chi(a)a^{k-1}a\left(\frac{nm}{a^2}\right)\right)e(nz).$$

This is equivalent to the claim upon making the change of variables $a\mapsto d$. $\square$

Now that we know how the Hecke operators act on the Fourier coefficients of modular forms, we can prove composition relations about them. However, we will require a rather tedious lemma about divisor pairs.

Lemma 3.10. *Fix positive integers m and n and a nonnegative r. For any positive integer d, set*

$$M(d)=\left\{d=d_1d_2 : d_1\mid(r,m)\ \text{and}\ d_2\mid\left(\frac{rm}{d_1^2},n\right)\right\}.$$

Then

$$M(d)=\left\{d=d_3d_4 : d_3\mid(m,n)\ \text{and}\ d_4\mid\left(r,\frac{mn}{d_3^2}\right)\right\}.$$

Proof. As the greatest common divisor is multiplicative in every entry, it suffice to prove the claim when $m=p^\alpha$, $n=p^\beta$, $r=p^\gamma$, and $d=d^\delta$ are all powers of a prime p. Then $d_1=p^{i_1}$ and $d_2=p^{i_2}$ with

$$0\leq i_1\leq\min(\alpha,\gamma)\quad\text{and}\quad 0\leq i_2\leq\min(\alpha+\gamma-2i_1,\beta),$$

while $d_3=p^{i_3}$ and $d_4=p^{i_4}$ with

$$0\leq i_2\leq\min(\alpha+\gamma-2i_1,\beta)\quad\text{and}\quad 0\leq i_4\leq\min(\alpha+\beta-2i_3,\gamma).$$

Moreover, $i_2=\delta-i_i$ and $i_4=\delta-i_3$. Substituting in this relations, our conditions become

$$\max(0,\delta-\beta)\leq i_1\leq\min(\alpha,\gamma,\delta,\alpha+\gamma-\delta),$$

and

$$\max(0,\delta-\gamma)\leq i_3\leq\min(\alpha,\beta,\delta,\alpha+\beta-\delta).$$

So our claim about $M(d)$ is proved if these intervals are of the same length. First suppose $\delta \leq \alpha$. Then the upper bounds are $\min(\gamma, \delta)$ and $\min(\beta, \delta)$ respectively. Therefore these intervals are of length

$$\min(\gamma, \delta) - \max(0, \delta - \beta) + 1 \quad \text{and} \quad \min(\beta, \delta) - \max(0, \delta - \gamma) + 1,$$

upon which a short computation shows are equal in every case. Now suppose $\delta > \alpha$. Then another short computation shows

$$\min(\alpha, \gamma, \delta, \alpha + \gamma - \delta) = \alpha - \max(0, \delta - \gamma) \quad \text{and} \quad \min(\alpha, \beta, \delta, \alpha + \beta - \delta) = \alpha - \max(0, \delta - \beta).$$

Whence these intervals are of length

$$\alpha - \max(0, \delta - \gamma) - \max(0, \delta - \beta) + 1 \quad \text{and} \quad \alpha - \max(0, \delta - \beta) - \max(0, \delta - \gamma) + 1,$$

which are identical. Therefore the claim is proved. □

With this lemma in hand, we can now prove the composition formula for the Hecke operators. This result will also show that the Hecke operators commute.

Theorem 3.11. *For any positive integers m and n, the Hecke operators T_m and T_n on $\mathcal{M}_k(\Gamma_1(N))$ satisfy*

$$T_m T_n = \sum_{d \mid (m,n)} \langle d \rangle d^{k-1} T_{\frac{mn}{d^2}} \quad \text{and} \quad T_{mn} = \sum_{d \mid (m,n)} \mu(d) \langle d \rangle d^{k-1} T_{\frac{m}{d}} T_{\frac{n}{d}},$$

are multiplicative, commutative, and also satisfy

$$T_p T_{p^r} = T_{p^{r+1}} + \langle p \rangle p^{k-1} T_{p^{r-1}},$$

for all primes p and positive integers r. In particular, on $\mathcal{M}_k(N, \chi)$ these Hecke operators satisfy

$$T_m T_n = \sum_{d \mid (m,n)} \chi(d) d^{k-1} T_{\frac{mn}{d^2}} \quad \text{and} \quad T_{mn} = \sum_{d \mid (m,n)} \mu(d) \chi(d) d^{k-1} T_{\frac{m}{d}} T_{\frac{n}{d}},$$

are multiplicative, commutative, and also satisfy

$$T_p T_{p^r} = T_{p^{r+1}} + \chi(p) p^{k-1} T_{p^{r-1}}.$$

Proof. In view of Proposition 3.6, it suffices to prove the statement for $\mathcal{M}_k(N, \chi)$. For the first identity, we will show that left-hand and right-hand sides act the same on the Fourier coefficients of any $f \in \mathcal{M}_k(N, \chi)$. So let

$$f(z) = \sum_{r \geq 0} a(r) e(rz),$$

be the Fourier series expansion of f. From two applications of Proposition 3.9, we find that

$$(T_m T_n f)(z) = \sum_{r \geq 0} \left(\sum_{\substack{d' \mid (r,m) \\ e' \mid \left(\frac{rm}{(d')^2}, n\right)}} \chi(d'e')(d'e')^{k-1} a\left(\frac{rmn}{(d'e')^2}\right) \right) e(rz).$$

By Lemma 3.10, we may write the right-hand side as

$$\sum_{r\geq 0}\left(\sum_{\substack{d\mid(m,n)\\ e\mid\left(r,\frac{mn}{d^2}\right)}}\chi(de)(de)^{k-1}a\left(\frac{rmn}{(de)^2}\right)\right)e(rz).$$

Another application of Proposition 3.9 shows that for fixed d the sum over e is exactly the r-th Fourier coefficient of $T_{\frac{mn}{d^2}}f$. Whence this previous expression is

$$\sum_{d\mid(m,n)}\chi(d)d^{k-1}\left(T_{\frac{mn}{d^2}}f\right)(z),$$

proving the first identity. To prove the second identity, fix a $d\mid(m,n)$ and write the first identity in the form

$$T_{\frac{m}{d}}T_{\frac{n}{d}}\sum_{e\mid\left(\frac{m}{d},\frac{n}{d}\right)}\chi(e)e^{k-1}T_{\frac{mn}{(de)^2}}.$$

Multiply each term by $\mu(d)\chi(d)d^{k-1}$ and sum over $d\mid(m,n)$ to obtain

$$\sum_{d\mid(m,n)}\mu(d)\chi(d)d^{k-1}T_{\frac{m}{d}}T_{\frac{n}{d}}=\sum_{d\mid(m,n)}\sum_{e\mid\left(\frac{m}{d},\frac{n}{d}\right)}\mu(d)\chi(de)(de)^{k-1}T_{\frac{mn}{(de)^2}}.$$

We now must show that the right-hand side is T_{mn}. The change of variables $e\mapsto\frac{e}{d}$ gives

$$\sum_{e\mid(m,n)}\sum_{d\mid e}\mu(d)\chi(e)e^{k-1}T_{\frac{mn}{e^2}}.$$

But the sum over d is

$$\sum_{d\mid e}\mu(d)=\begin{cases}1 & \text{if } e=1,\\ 0 & \text{if } e\geq 2.\end{cases}$$

Therefore every term vanishes except the $e=1$ term which is exactly T_{mn}. This proves the second identity.

Multiplicativity follows from either formula since we only have the term $d=1$ whenever $(m,n)=1$. As for commutativity, just observe that either formula is symmetric in m and n. The formula is immediate from either of the previous ones as $d=1,p$ when $m=p$ and $n=p^r$. $\square$

We can now prove that the diamond and Hecke operators $\langle m\rangle$ and T_m are normal whenever $(m,N)=1$. We first note two important consequences from Equations (11) and (12) which is that in the case $m=p$ for a prime p, we have

$$(T_pf)(z)=\left(\left[\Gamma_1(N)\begin{pmatrix}1&0\\0&p\end{pmatrix}\Gamma_1(N)\right]_kf\right)(z), \tag{14}$$

and

$$(T_p f)(z) = \sum_{b \pmod p} \left(f \bigg\|_k \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix} \right)(z) + \left((\langle p \rangle f) \bigg\|_k \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \right)(z). \tag{15}$$

In particular, we find that the Hecke operator T_p reduces to the single double coset operator $\left[\Gamma_1(N) \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \Gamma_1(N)\right]_k$. Now we will compute the aforementioned adjoints explicitly.

Theorem 3.12. *Let m be a positive integer with $(m, N) = 1$ and m^* be any positive integer satisfying $m^* m \equiv 1 \pmod N$. The adjoints of the diamond and Hecke operators $\langle m \rangle$ and T_m on $\mathcal{S}_k(\Gamma_1(N))$ are given by*

$$\langle m \rangle^* = \langle m^* \rangle \quad \text{and} \quad T_m^* = \langle m^* \rangle T_m,$$

and they are are normal. In particular, on $\mathcal{S}_k(N, \chi)$ the adjoints of these operators are given by

$$\langle m \rangle^* = \overline{\langle m \rangle} \quad \text{and} \quad T_m^* = \overline{\langle m \rangle} T_m,$$

and they are also normal.

Proof. By Theorem 3.11 and induction, it suffices to prove the adjoint formulas when $m = p$ for a prime p with $p \nmid N$. We will prove the adjoint formula for diamond operators first. For any γ_p, we have $\gamma_p^* \in \Gamma_0(N)$ with $\gamma_p^* \equiv \left(\begin{smallmatrix} p & 0 \\ 0 & p^* \end{smallmatrix}\right) \pmod N$. Therefore Proposition 2.13 implies

$$\langle p \rangle^* = \langle p^* \rangle.$$

This proves the adjoint formula for the diamond operators and normality follows from commutativity. For the Hecke operators, notice that as $\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)^* = \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$ we may write $\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)^* = \gamma_{p^*} \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)$. So from Equation (14) and Proposition 2.13 we have

$$T_p = \left[\Gamma_1(N) \gamma_{p^*} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma_1(N) \right]_k.$$

By normality of $\Gamma_1(N)$ in $\Gamma_0(N)$, it follows that

$$T_p = \langle p^* \rangle T_p.$$

This proves the adjoint formula for the Hecke operators and normality follows from commutativity.

In the case of $\mathcal{S}_k(N, \chi)$, the formulas follow from the observation that $\chi(m^*) = \overline{\chi}(m)$ and normality again follows from commutativity. $\square$

Remark 3.13. Note that on $\mathcal{S}_k(\mathrm{SL}_2(\mathbb{Z}))$, all of the diamond operators are the identity operator and so they are self-adjoint. Moreover, this implies $T_p^* = T_p$ for all primes p so that the Hecke operators are self-adjoint as well.

Let $f \in \mathcal{S}_k(N, \chi)$ be non-constant. Then f is already an eigenfunction for the diamond operators $\langle d \rangle$ with eigenvalue $\chi(d)$. If f is also an eigenfunction for the Hecke operator T_m with eigenvalue $\lambda(m)$, so that

$$(T_m f)(z) = \lambda(m) f(z),$$

we call $\lambda(m)$ the m-th **Hecke eigenvalue** of f. If f is a simultaneous eigenfunction for all Hecke operators T_m with $(m,N)=1$, we call f an **eigenform**. If the condition $(m,N)=1$ can be dropped, so that f is a simultaneous eigenfunction for all the Hecke operators, we say f is a **Hecke eigenform**. In particular, on $\mathcal{S}_k(1)$ all eigenforms are Hecke eigenforms. If f is a Hecke eigenform with Fourier coefficients $a(n)$ then Proposition 3.9 immediately implies that the first Fourier coefficient of $T_m f$ is $a(m)$ and so

$$a(m)=\lambda(m)a(1).$$

Therefore we cannot have $a(1)=0$ for this would mean f is constant. So we can normalize f by dividing by $a(1)$ which guarantees that this Fourier coefficient is 1. This is called the **Hecke normalization** of f and it ensures

$$a(m)=\lambda(m).$$

The **Petersson normalization** of f is where we normalize so that $\langle f,f\rangle=1$. In particular, any orthonormal basis of $\mathcal{S}_k(\Gamma_1(N))$ is Petersson normalized.

In any case, we now have developed enough about the Hecke operators to show that we can always find a basis of eigenforms.

Theorem 3.14. *$\mathcal{S}_k(\Gamma_1(N))$ admits an orthonormal basis of eigenforms. In particular, so does $\mathcal{S}_k(N,\chi)$.*

Proof. In view of Proposition 3.6, it suffices to prove the latter statement. By Theorems 2.7, 3.11 and 3.12, the Hecke operators T_m with $(m,N)=1$ are a commuting family of normal operators on the finite dimensional complex Hilbert space $\mathcal{S}_k(N,\chi)$. Therefore they admit a simultaneous orthonormal eigenbasis by the spectral theorem. This basis is exactly an orthonormal basis of eigenforms. $\square$

The Hecke eigenvalues of Hecke eigenforms satisfy certain relations known as the **Hecke relations** for modular forms.

Proposition (Hecke relations). *Let $f\in\mathcal{S}_k(N,\chi)$ be an eigenform with Hecke eigenvalues $\lambda(m)$. Then the Hecke eigenvalues are multiplicative and satisfy*

$$\lambda(m)\lambda(n)=\sum_{d\mid(m,n)}\chi(d)d^{k-1}\lambda\left(\frac{mn}{d^2}\right)\quad \text{and}\quad \lambda(mn)=\sum_{d\mid(m,n)}\mu(d)\chi(d)d^{k-1}\lambda\left(\frac{m}{d}\right)\lambda\left(\frac{n}{d}\right),$$

for all positive integers m and n with $(mn,N)=1$. If f is a Hecke eigenform, these relations hold for all m and n.

Proof. First let f be an eigenform and, if necessary, Hecke normalize f. Multiplicativity of the Hecke eigenvalues now follows from that of the Hecke operators. To prove the first identity, we will compute the first Fourier coefficient of T_mT_nf in two ways. On the one hand, since f is a Hecke normalized eigenform it is $\lambda(n)\lambda(m)$. On the other hand, it is $\sum_{d\mid(n,m)}\chi(d)d^{k-1}\lambda\left(\frac{nm}{d^2}\right)$ by Theorem 3.11. The second identity is proved analogously by computing the fist Fourier coefficient of $T_{mn}f$ in two ways. On the one hand it is $\lambda(mn)$ since f is a Hecke normalized eigenform. On the other hand, it is $\sum_{d\mid(m,n)}\mu(d)\chi(d)d^{k-1}\lambda\left(\frac{m}{d}\right)\lambda\left(\frac{n}{d}\right)$ by Theorem 3.11. This proves the first statement. The second statement is clear since in that case f is an eigenfunction for all the Hecke operators. $\square$

3.3 Todo: [Oldforms and Newforms]

Recall that if $M \mid N$ then $\Gamma_1(N) \subseteq \Gamma_1(M)$ and so every modular form on $\Gamma_1(M)\backslash\mathbb{H}$ is also a modular form on $\Gamma_1(N)\backslash\mathbb{H}$. In particular, there are inclusions

$$\mathcal{M}_k(\Gamma_1(M)) \subseteq \mathcal{M}_k(\Gamma_1(N)) \quad \text{and} \quad \mathcal{S}_k(\Gamma_1(M)) \subseteq \mathcal{S}_k(\Gamma_1(N)).$$

The purpose of introducing oldforms and newforms will allow us to distinguish which modular forms on a modular curve are induced from those on a modular curve of lower level and which are not. The former are referred to as oldforms and the latter newforms. Our primary goal will be to investigate how the Hecke operators act on the subspace generated by the newforms for which we can say more than the oldforms.

As just remarked, inclusion is the easiest way to lift a modular form from $\Gamma_1(M)\backslash\mathbb{H}$ to $\Gamma_1(N)\backslash\mathbb{H}$. However, there is a less obvious way of lifting. For any $d \mid \frac{N}{M}$, let $\alpha_d = \left(\begin{smallmatrix} d & 0 \\ 0 & 1 \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{Q})$ and consider the double slash operator $\|_k\alpha_d : C(\mathbb{H}) \to C(\mathbb{H})$ given by

$$(f\|_k\alpha_d)(z) = d^{k-1}f(dz).$$

It turns out that this operator lifts modular forms from $\Gamma_1(M)\backslash\mathbb{H}$ to $\Gamma_1(N)\backslash\mathbb{H}$ and preserves the subspaces of cusp forms.

Proposition 3.15. *Let M and N be positive integers such that $M \mid N$. For any $d \mid \frac{N}{M}$, $\|_k\alpha_d$ maps $\mathcal{M}_k(\Gamma_1(M))$ into $\mathcal{M}_k(\Gamma_1(N))$ and preserves the subspaces of cusp forms.*

Proof. Let $f \in \mathcal{M}_k(\Gamma_1(M))$. It is clear that holomorphy is satisfied for $f\|_k\alpha_d$. To verify modularity, let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d' \end{smallmatrix}\right) \in \Gamma_1(N)$. A short computation shows

$$\alpha_d\gamma\alpha_d^{-1} = \begin{pmatrix} a & bd \\ d^{-1}c & d' \end{pmatrix},$$

whence $\alpha_d\gamma\alpha_d^{-1} \in \Gamma_1(M)$. But then $\alpha_d\Gamma_1(N)\alpha_d^{-1} \subseteq \Gamma_1(M)$ which is to say $\Gamma_1(N) \subseteq \alpha_d^{-1}\Gamma_1(M)\alpha_d$. Modularity then follows from a short computation. For the growth condition, let $\sigma_{\mathfrak{a}}$ be a scaling matrix for a cusp $\mathfrak{a}$ of $\Gamma_1(M)\backslash\mathbb{H}$. Then $\alpha_d\sigma_{\mathfrak{a}}\infty = \mathfrak{b}$ for some cusp $\mathfrak{b}$ of $\Gamma_1(N)\backslash\mathbb{H}$ and as

$$((f\|_k\alpha_d)|_k\sigma_{\mathfrak{a}})(z) = d^{\frac{k}{2}-1}(f|_k\alpha_d\sigma_{\mathfrak{a}})(z),$$

the growth condition follows from that of f. This proves the first statement. The second statement follows from this identity since $f\|_k\alpha_d$ is a cusp form if f is. $\square$

We can now define oldforms and newforms. For each divisor d of N, consider the bilinear operator

$$i_d : \mathcal{S}_k\left(\Gamma_1\left(\frac{N}{d}\right)\right) \oplus \mathcal{S}_k\left(\Gamma_1\left(\frac{N}{d}\right)\right) \to \mathcal{S}_k(\Gamma_1(N)) \qquad (f, g) \mapsto f + g\|_k\alpha_d.$$

This map is well-defined by Proposition 3.15. The subspaces of **oldforms** and **newforms** of level N are given by

$$\mathcal{S}_k^{\mathrm{old}}(\Gamma_1(N)) = \bigoplus_{p \mid N} \mathrm{Im}(i_p) \quad \text{and} \quad \mathcal{S}_k^{\mathrm{new}}(\Gamma_1(N)) = \mathcal{S}_k^{\mathrm{old}}(\Gamma_1(N))^{\perp}.$$

An element of these subspaces is called an **oldform** or **newform** respectively. Note that there are no oldforms of level 1. We will need a useful operator for the study of oldforms and newforms. Let

$$W_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix},$$

and note that $\det(W_N) = N$. This matrix normalizes $\Gamma_0(N)$. Indeed, let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$. A short computation shows

$$W_N \gamma W_N^{-1} = \begin{pmatrix} d & -N^{-1}c \\ -Nb & a \end{pmatrix},$$

which clearly belongs to $\Gamma_0(N)$. So W_N normalizes $\Gamma_0(N)$ as claimed. In particular, we will set $\gamma_W = W_N \gamma W_N^{-1} \in \Gamma_0(N)$ so that the previous identity can be expressed as

$$W_N \gamma = \gamma_W W_N. \tag{16}$$

We define the **Atkin-Lehner operator** $\omega_N : C(\mathbb{H}) \to C(\mathbb{H})$ to be the linear operator given by

$$(\omega_N f)(z) = N^{1-k}(f\|_k W_N)(z).$$

As W_N is invertible so is the Atkin-Lehner operator ω_N. It is not difficult to see how the Atkin-Lehner operator acts on $\mathcal{S}_k(\Gamma_1(N))$.

Proposition 3.16. *The Atkin-Lehner operator ω_N maps $\mathcal{S}_k(\Gamma_1(N))$ into itself. In particular, ω_N takes $\mathcal{S}_k(N,\chi)$ into $\mathcal{S}_k(N,\overline{\chi})$ and preserves the subspaces of oldforms and newforms. Moreover, the adjoint is given by*

$$\omega_N^* = (-1)^k \omega_N,$$

and we also have

$$\omega_N^2 = (-1)^k.$$

Proof. In light of Proposition 3.6, the first statement is a consequence of the latter ones. So let $f \in \mathcal{S}_k(N,\chi)$. Holomorphy of $\omega_N f$ is obvious. For modularity, let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$ and recall $\gamma_W \in \Gamma_0(N)$. Then $\chi(\gamma_W) = \overline{\chi}(\gamma)$ and modularity follows from a short computation using Equation (16). As for the growth condition, let $\sigma_{\mathfrak{a}}$ be a scaling matrix for a cusp $\mathfrak{a}$. Then $W_N \sigma_{\mathfrak{a}} \infty = \mathfrak{b}$ for some cusp $\mathfrak{b}$ of $\Gamma_1(N)\backslash\mathbb{H}$. As

$$((\omega_N f)|_k \sigma_{\mathfrak{a}})(z) = N^{-\frac{k}{2}}(f|_k W_N \sigma_{\mathfrak{a}})(z),$$

the growth condition follows from that of f. In particular, $\omega_N f$ is a cusp form because f. This proves ω_N takes $\mathcal{S}_k(N,\chi)$ into $\mathcal{S}_k(N,\overline{\chi})$. We now show the adjoint formula for the Atkin-Lehner operator. First observe that the adjugate of W_N satisfies $W_N^* = -W_N$. Therefore Proposition 2.13 implies

$$\omega_N^* = (-1)^k \omega_N,$$

proving the desired adjoint formula. With the adjoint formula, we can further show that the Atkin-Lehner operator preserves the subspaces of oldforms and newforms. To show that it preserves the subspace of oldforms, let p be a prime satisfying $p \mid N$ and suppose $i_p(f,g) = h$ so that $h = f + g\|_k \alpha_p$. Then it is enough to show

$$\omega_N i_p(f,g) = i_p(\omega_{\frac{N}{p}} p^{k-2} g, \omega_{\frac{N}{p}} p^{1-k} f),$$

which will follow from the formulas

$$\omega_N f = (\omega_{\frac{N}{p}} p^{1-k} f)\|_k \alpha_p \quad \text{and} \quad \omega_N(g\|_k \alpha_p) = \omega_{\frac{N}{p}}(p^{k-2} g).$$

These formulas are implied by the identities

$$W_N = W_{\frac{N}{p}} \alpha_p \quad \text{and} \quad \alpha_p W_N = I_p W_{\frac{N}{p}}.$$

Thus the Atkin-Lehner operator ω_N preserves the subspace of oldforms. To see that ω_N preserves the subspace of newforms as well, let f and g be a newform and an oldform respectively. The adjoint formula for ω_N gives

$$\langle \omega_N f, g \rangle = \langle f, (-1)^k \omega_N g \rangle.$$

But the right-hand side is zero because it is an inner product between a newform and an oldform. Hence $\omega_N f$ must also be a newform and so ω_N preserves the subspace of newforms. It remains to prove the last formula. For this, observe that $W_N^2 = -I_N$. Using this fact, a short computation shows

$$(\omega_N^2 f)(z) = (-1)^k f.$$

This completes the proof. □

This result showed that the Atkin-Lehner operator ω_N is self-adjoint and an involution whenever k is even. In fact, regardless of the parity of k, ω_N is at most of order 4. We now need to understand how the Atkin-Lehner operator interacts with the diamond and Hecke operators. It turns out that the adjoints of these operators are given by conjugation via the Atkin-Lehner operator.

Proposition 3.17. *On $\mathcal{S}_k(\Gamma_1(N))$ and for all positive integers m, the diamond and Hecke operators $\langle m \rangle$ and T_m satisfy the adjoint formulas*

$$\langle m \rangle^* = \omega_N \langle m \rangle \omega_N^{-1} \quad \textit{and} \quad T_m^* = \omega_N T_m \omega_N^{-1}.$$

Proof. As the diamond and Hecke operators are multiplicative, it suffices to prove the two adjoint formulas when $m = p$ for a prime p. For the diamond operators, the formula is obvious when $p \mid N$ since $\langle p \rangle$ is the zero operator. So suppose $p \nmid N$ and observe from Equation (16) that $\gamma_p^* = W_N \gamma_p W_N^{-1}$. Whence Proposition 2.13 gives

$$\langle p \rangle^* = \omega_N \langle p \rangle \omega_N^{-1}.$$

This is the desired adjoint formula for the diamond operators when $p \mid N$. Thus the adjoint formula holds for all the diamond operators. For the Hecke operators, Equation (16) shows that $\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)^* = W_N \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) W_N^{-1}$. Then Equation (14) and Proposition 2.13 together imply

$$T_p^* = \omega_N T_p \omega_N^{-1}.$$

This is the desired adjoint formula for the Hecke operators. □

It also turns out that the spaces of oldforms and newforms are invariant under the diamond and Hecke operators.

Proposition 3.18. *On $\mathcal{S}_k(\Gamma_1(N))$ and for all positive integers m, the diamond and Hecke operators $\langle m\rangle$ and T_m preserve the subspaces of oldforms and newforms.*

Proof. By multiplicativity of the diamond and Hecke operators, it suffices to prove this when $m = p$ for a prime p. Moreover, if $N = 1$ the result is trivial because there are no oldforms. We will first show that the diamond and Hecke operators preserve the subspace of oldforms. So let q be a prime satisfying $q \mid N$ and suppose $i_q(f, g) = h$ which is to say $h = f + g\|_k\alpha_q$. For the diamond operators, the result is trivial if $p \mid N$ since $\langle p\rangle i_q$ is the zero operator. If $p \nmid N$, it is enough to show

$$\langle p\rangle i_q(f, g) = i_q(\langle p\rangle f, \langle p\rangle g),$$

which will follow from the formulas

$$\langle p\rangle f = \langle p\rangle f \quad \text{and} \quad \langle p\rangle(g\|_k\alpha_q) = (\langle p\rangle g)\|_k\alpha_q,$$

where the diamond operators on the left-hand and right-hand sides are on levels N and $\frac{N}{q}$ respectively. As $p \nmid N$, we can choose $\gamma_p \in \Gamma_0(N) \cap \Gamma_0\left(\frac{N}{q}\right)$ and the first formula holds. The second formula follows from the identity

$$\begin{pmatrix} * & * \\ * & q \end{pmatrix}\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} * & * \\ * & q \end{pmatrix}.$$

Therefore the diamond operators preserve the subspace of oldforms. The situation is slightly more involved for the Hecke operators. In the case $p \neq q$, it is enough to show

$$T_p i_q(f, g) = i_q(T_p f, T_p g),$$

which will follow from the formulas

$$T_p f = T_p f \quad \text{and} \quad T_p(g\|_k\alpha_q) = (T_p g)\|_k\alpha_p,$$

where the Hecke operators on the left-hand and right-hand sides are on levels N and $\frac{N}{q}$ respectively. The first formula follows immediately from Proposition 3.9 as the action of these Hecke operators on Fourier coefficients is identical. For the second formula, we may assume $g \in \mathcal{S}\left(\frac{N}{p}, \chi\right)$ in view of Proposition 3.6. Then $g\|_k\alpha_p \in \mathcal{S}_k(N, \chi)$ from Proposition 3.15 and the second formula follows by comparing Fourier coefficients using Proposition 3.9. In the case $p = q$, it is enough to show

$$T_q i_q(f, g) = i_q(T_q f + q^{k-1}g, -\langle q\rangle f),$$

which will follow from the formulas

$$T_q f = T_q f - (\langle q\rangle f)\|_k\alpha_q \quad \text{and} \quad T_q(g\|_k\alpha_q) = q^{k-1}g,$$

where the diamond and Hecke operators on the left-hand and right-hand sides are on levels N and $\frac{N}{q}$ respectively. The first formula holds in view of Equation (15) since $\langle q\rangle$ is the zero operator if $q \mid \frac{N}{q}$ or it annihilates the last term in the Hecke operator if $q \nmid \frac{N}{q}$. Similarly, a short computation using Equation (15) gives the second identity. This shows that the second formula holds. Therefore the Hecke operators also preserve the subspace of oldforms. We now show that the diamond and Hecke operators also preserve the subspace of newforms. So let f and g be a newform and an oldform respectively. Then for any positive integer m, Proposition 3.17 gives

$$\langle \langle m \rangle f, g \rangle = \langle f, \omega_N \langle m \rangle \omega_N^{-1} g \rangle \quad \text{and} \quad \langle T_m f, g \rangle = \langle f, \omega_N T_m \omega_N^{-1} g \rangle.$$

In view of Proposition 3.16, the right-hand sides are zero because they are inner products between a newform and an oldform. Therefore $\langle m \rangle f$ and $T_m f$ are newforms which shows that the diamond and Hecke operators preserve the subspace of newforms. $\square$

As a corollary, the subspaces of oldforms and newforms admit orthonormal bases of eigenforms.

Corollary 3.19. *$\mathcal{S}_k^{\text{old}}(\Gamma_1(N))$ and $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$ admit orthonormal bases of eigenforms.*

Proof. This follows immediately from Theorem 3.14 and Proposition 3.18. $\square$